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Chang et al.(10) **Pub. No.: US 2025/0287577 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SEMICONDUCTOR DEVICES HAVING
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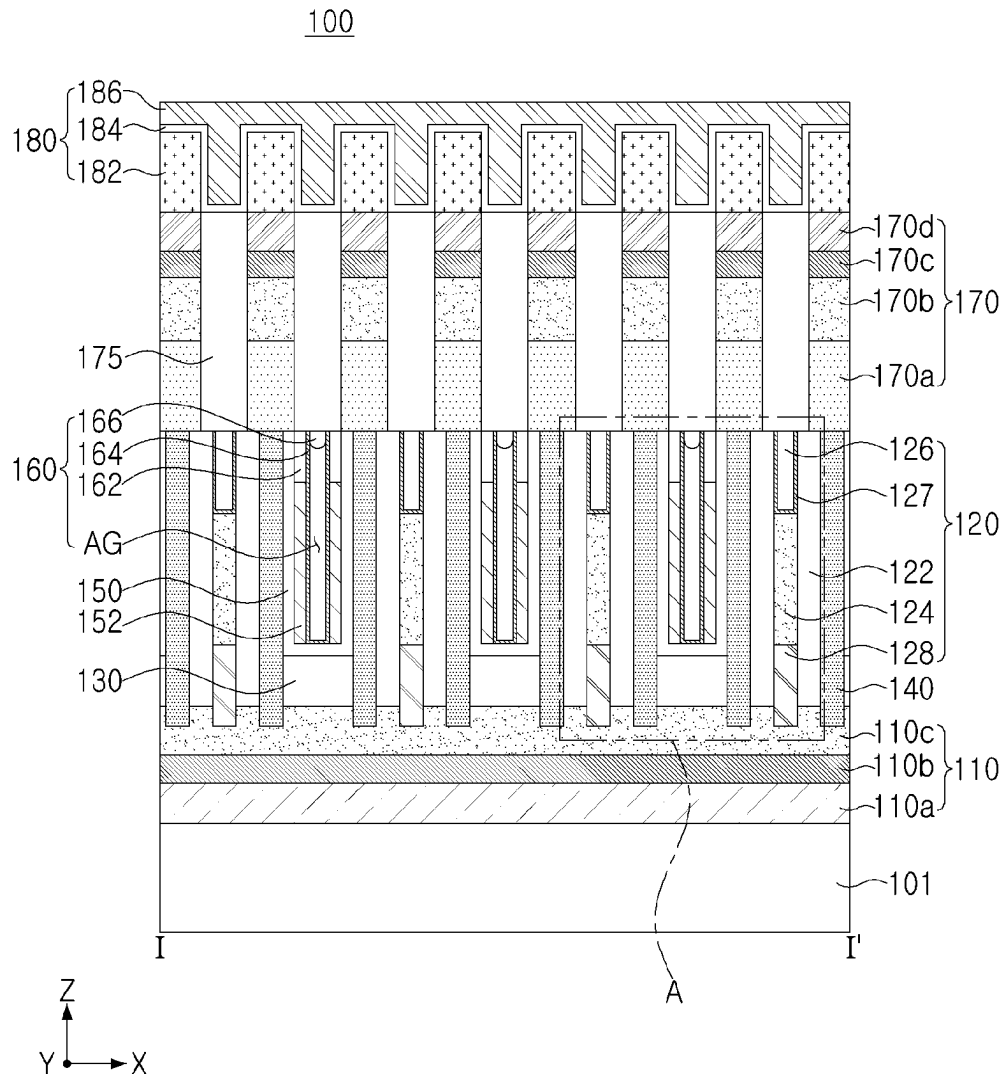
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(57)

ABSTRACT

A semiconductor device, comprising: a bit line structure; back gate structures on the bit line structure; a first channel layer and a second channel layer between the back gate structures; a first word line and a second word line between the first and second channel layers, wherein the first word line is adjacent the first channel layer, and the second word line is adjacent the second channel layer; and an insulating structure on side surfaces of the first and second word lines and upper surfaces of the first and second word lines, wherein the insulating structure includes a core region between the first and second word lines; and an insulating liner between the core region and the first and second word lines, wherein the side surfaces of the first and second word lines face each other, wherein the insulating liner extends to a level higher than those of the first and second word lines.



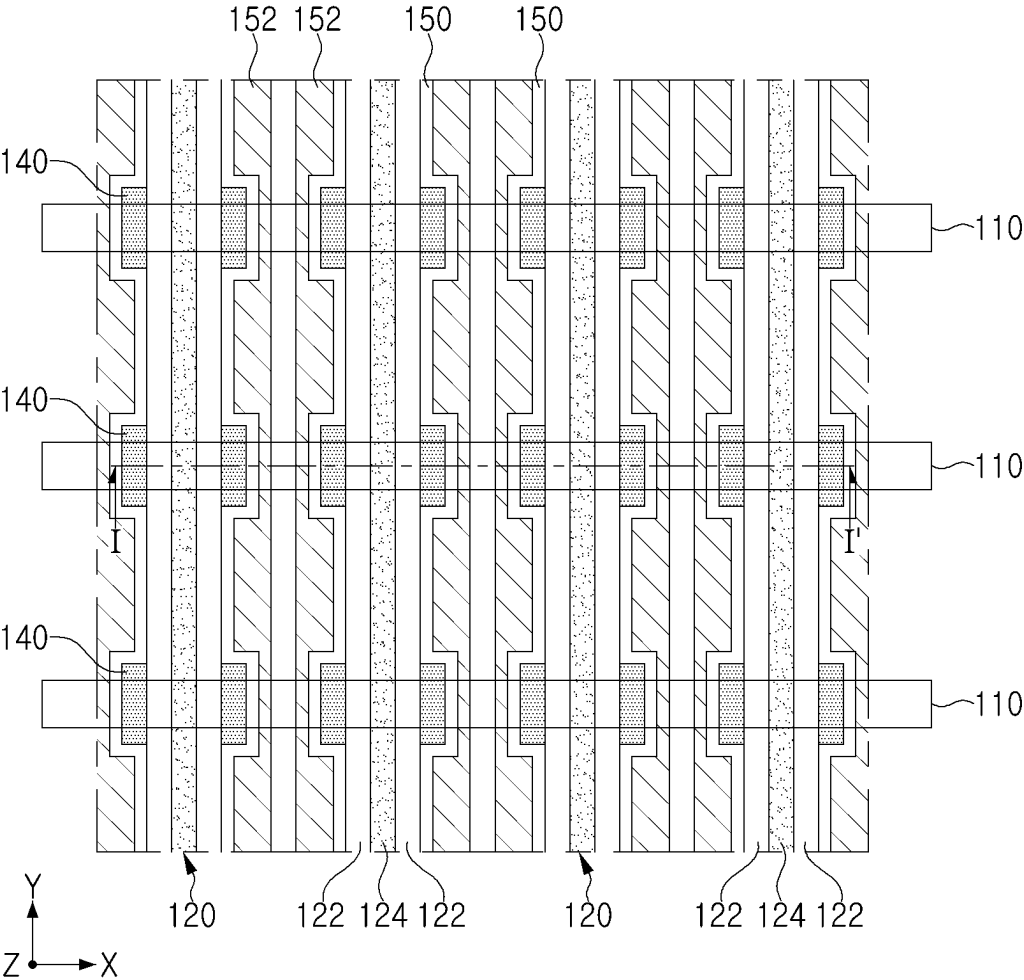


FIG. 1

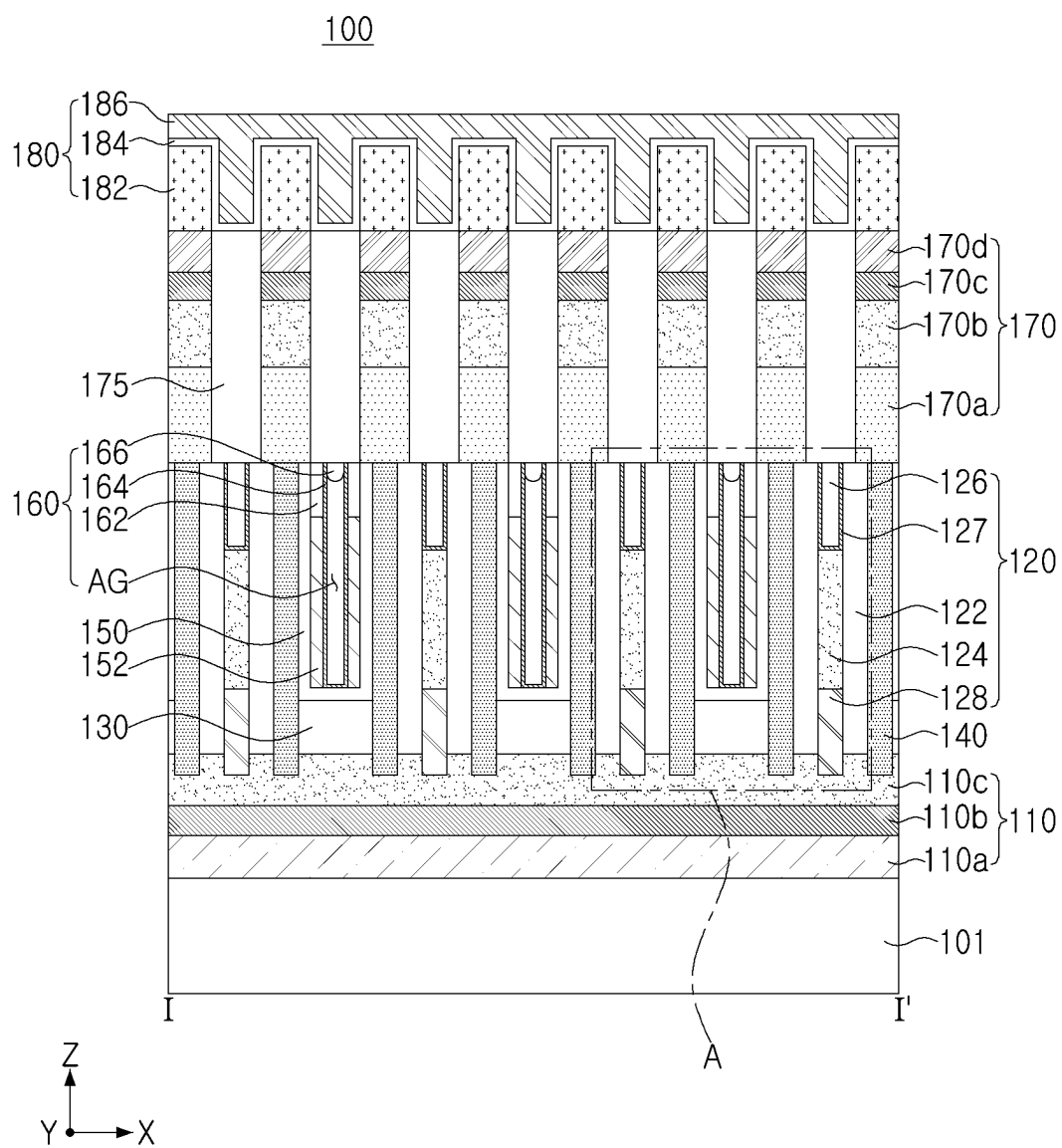


FIG. 2

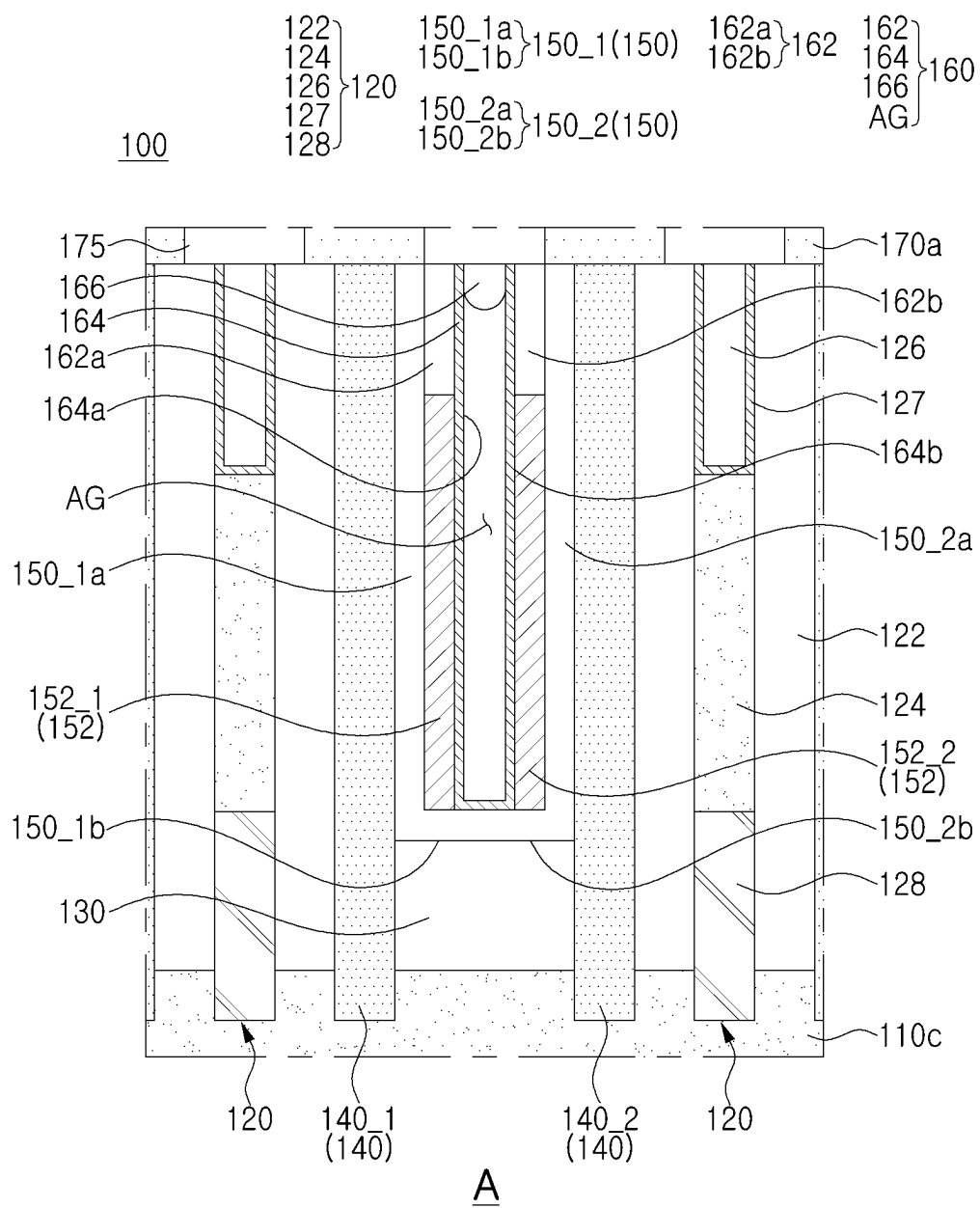


FIG. 3

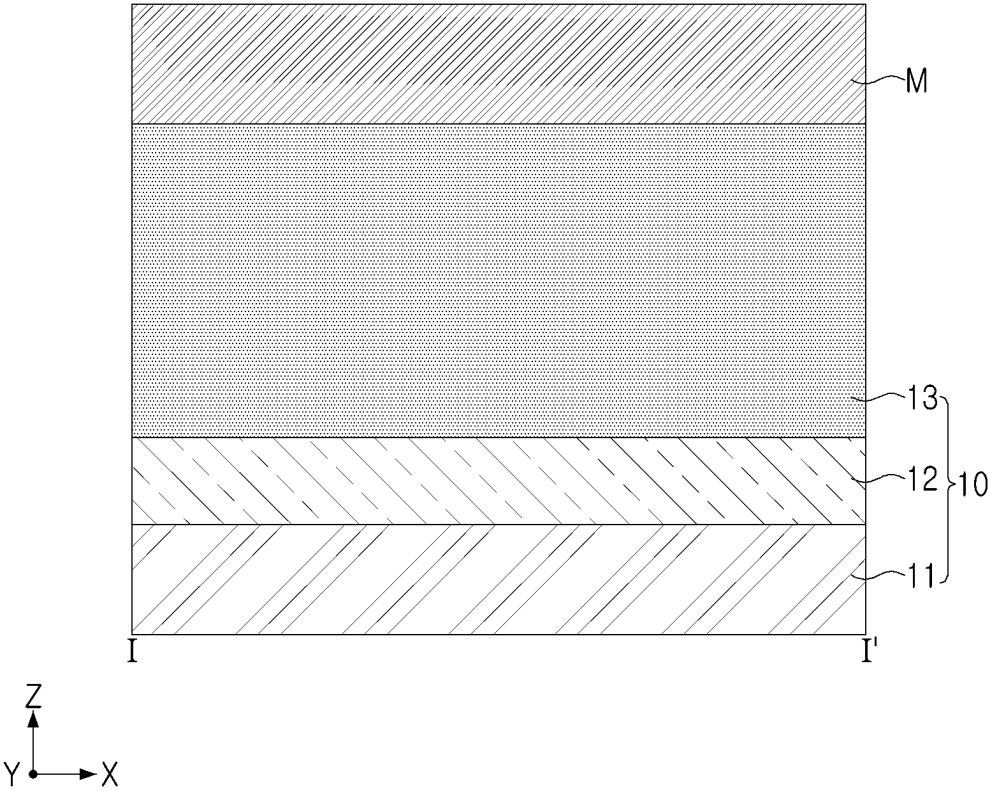


FIG. 4

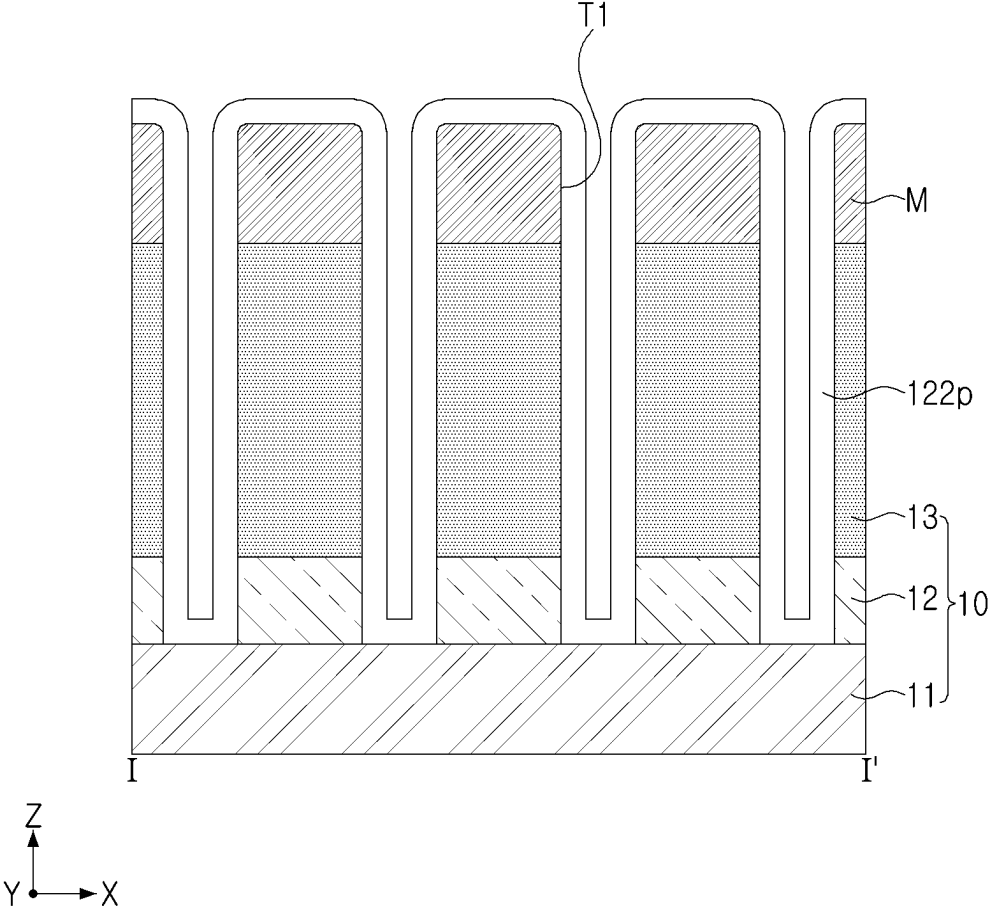


FIG. 5

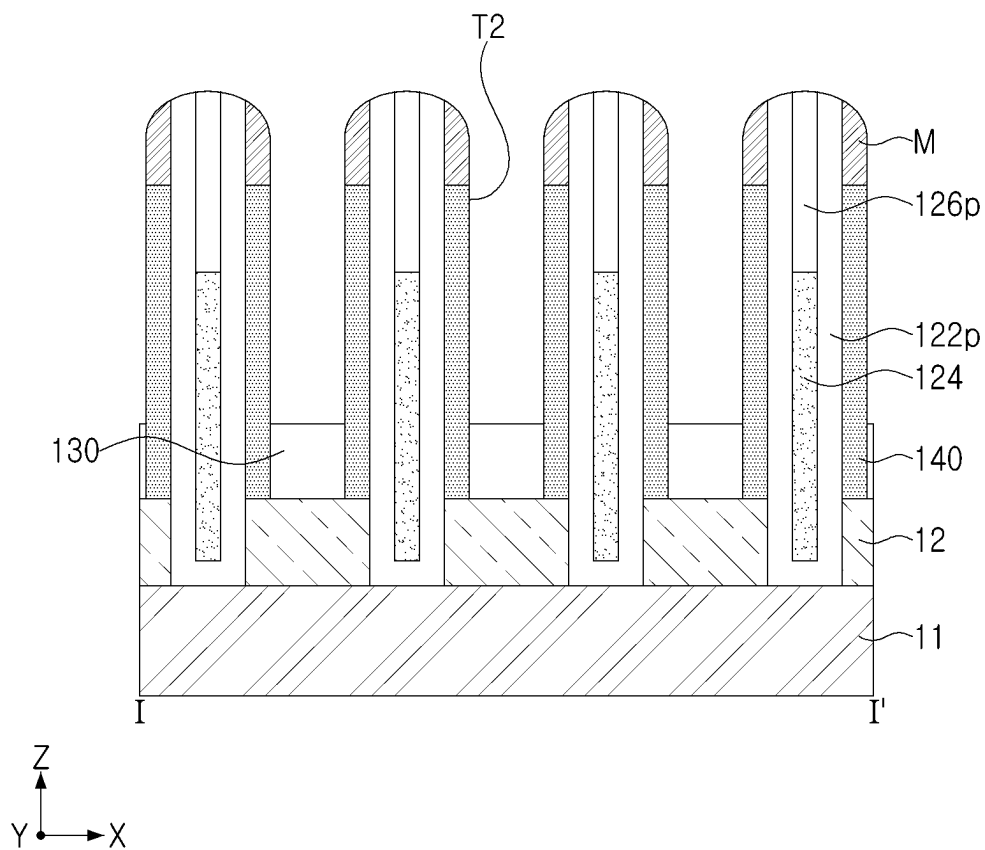


FIG. 7

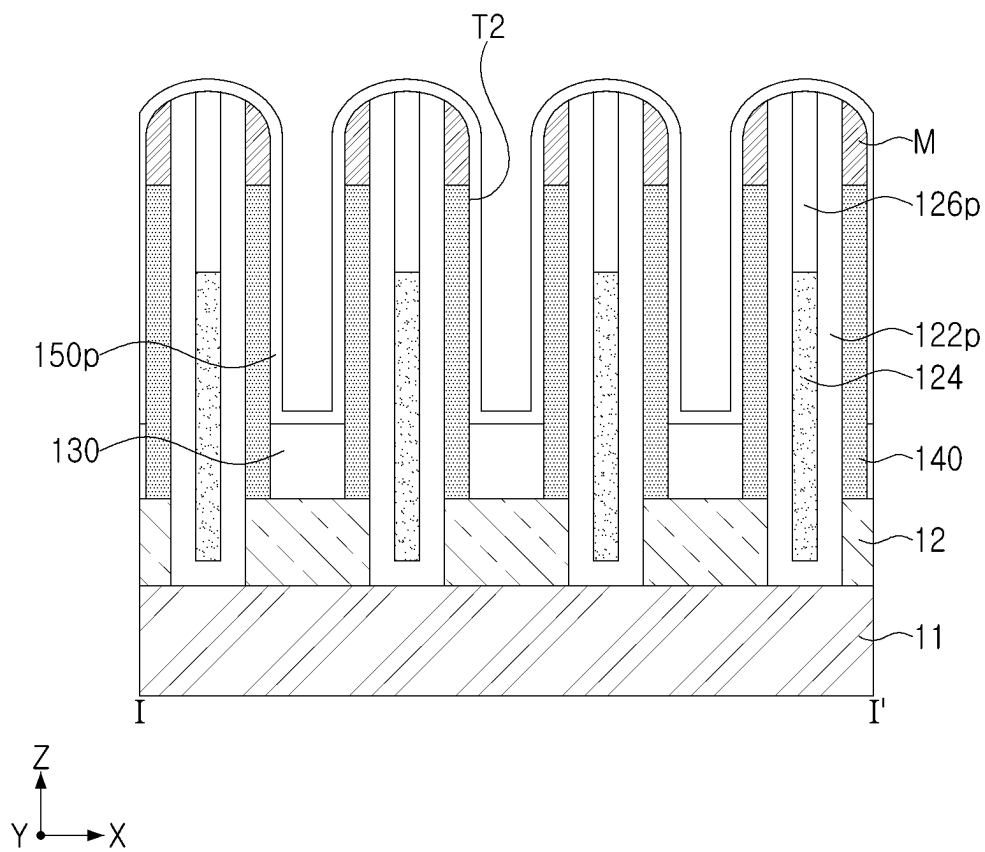


FIG. 8

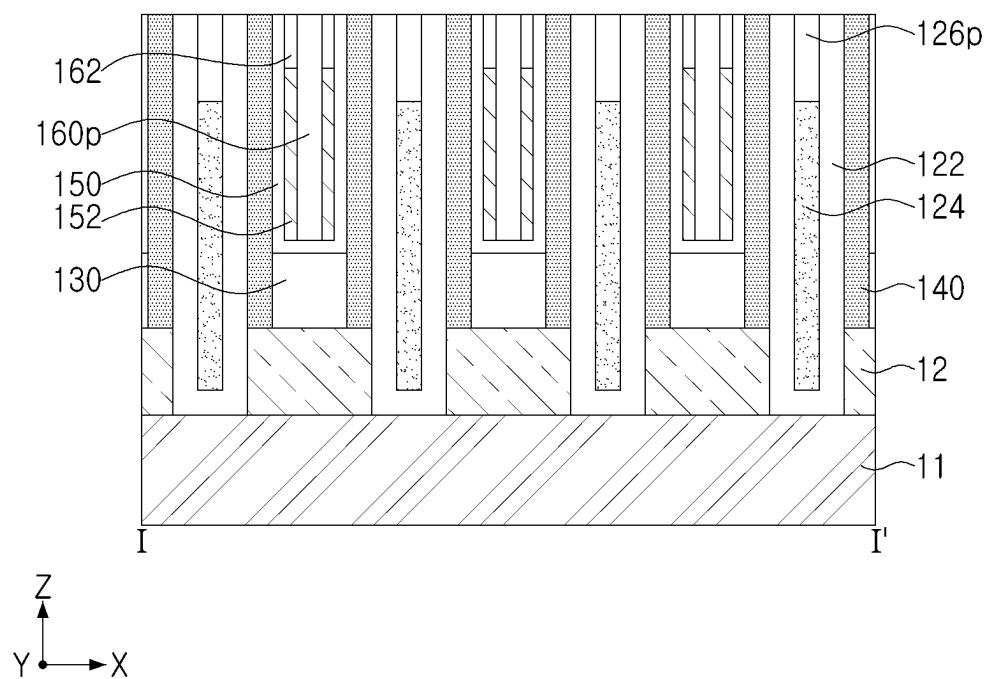


FIG. 9

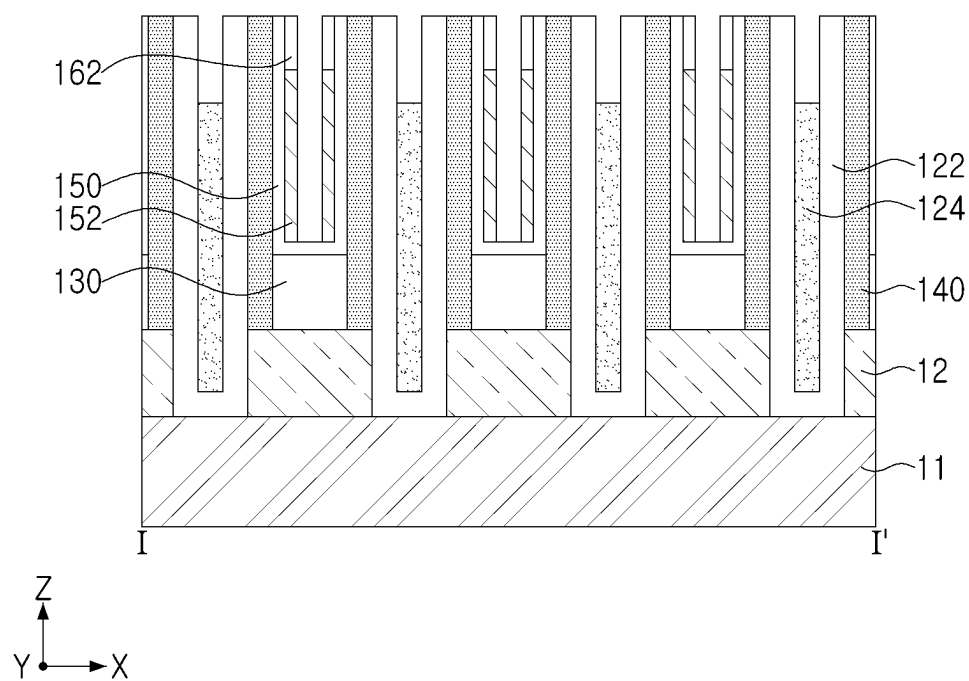


FIG. 10

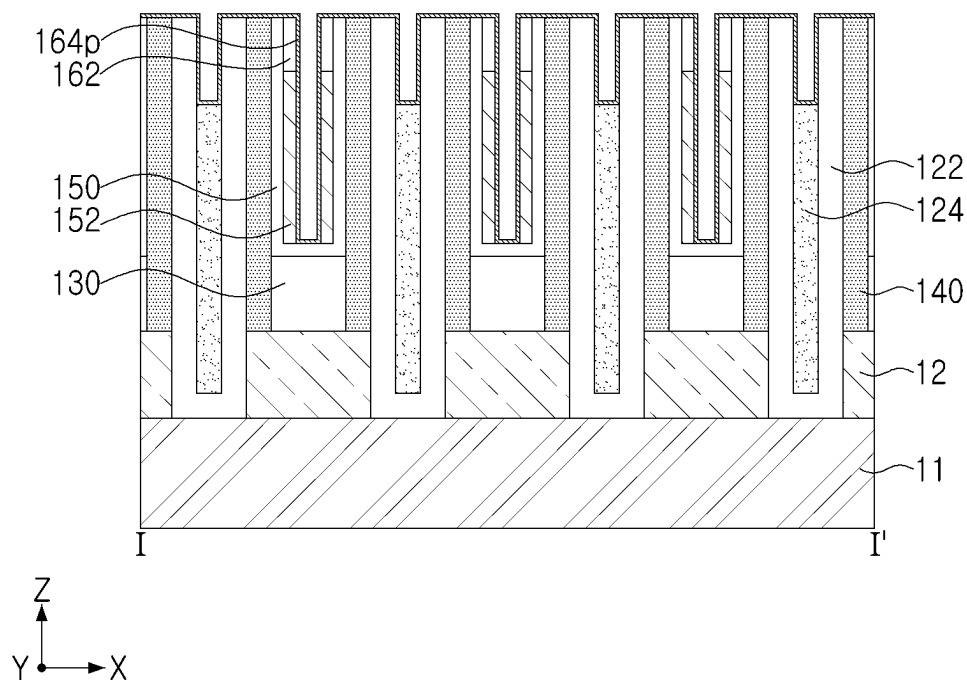


FIG. 11

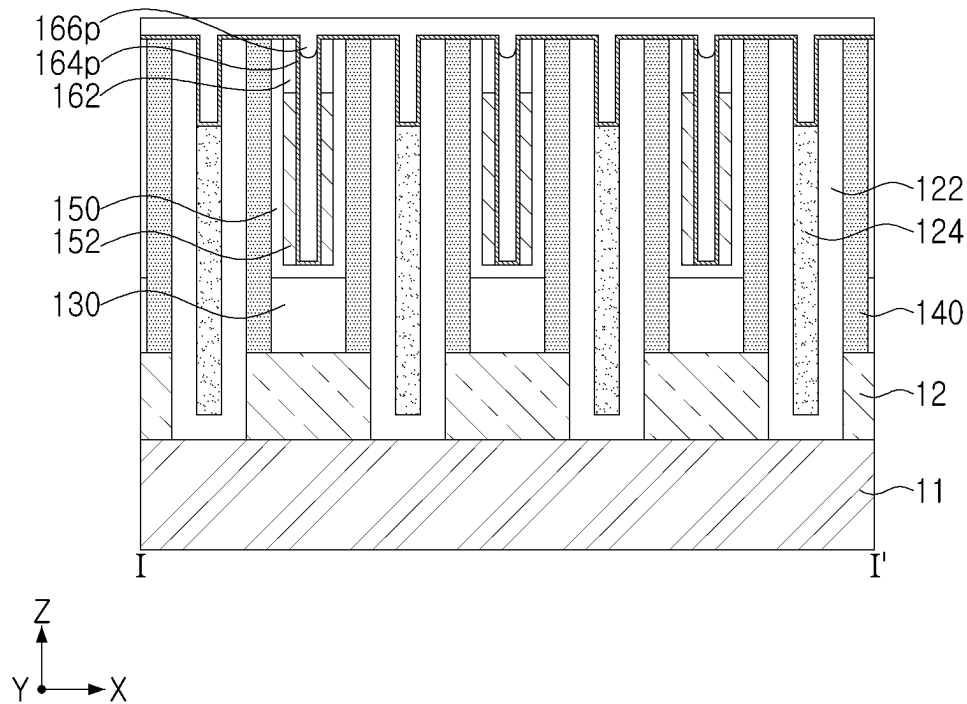
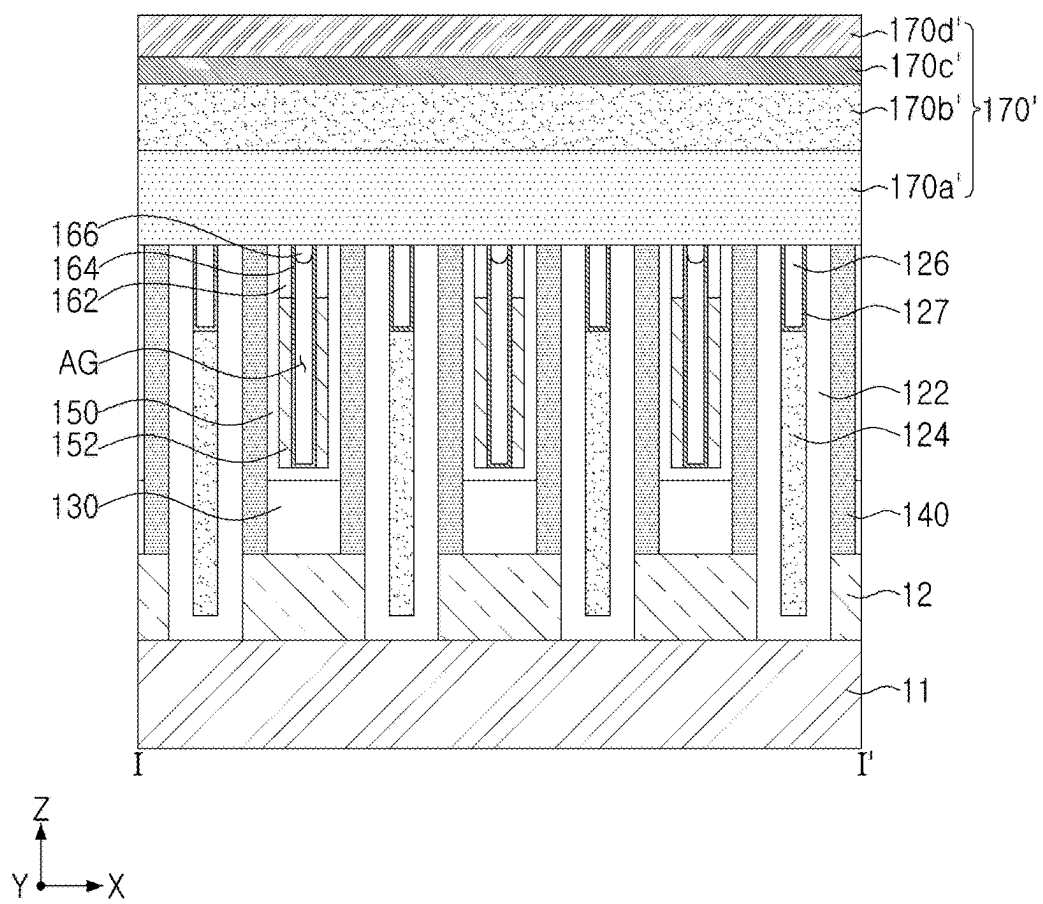
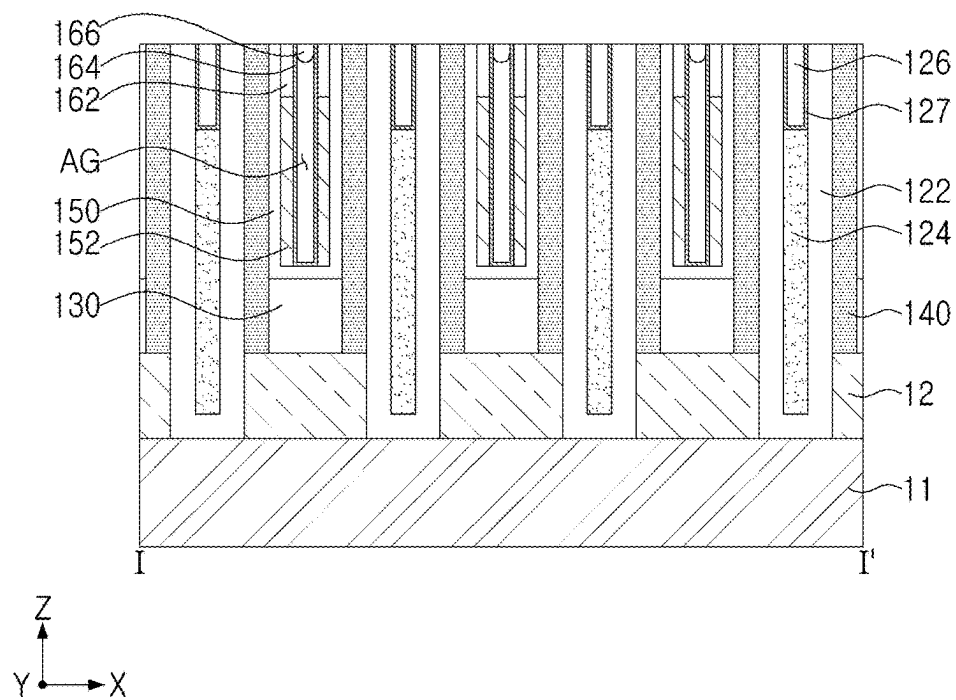


FIG. 12



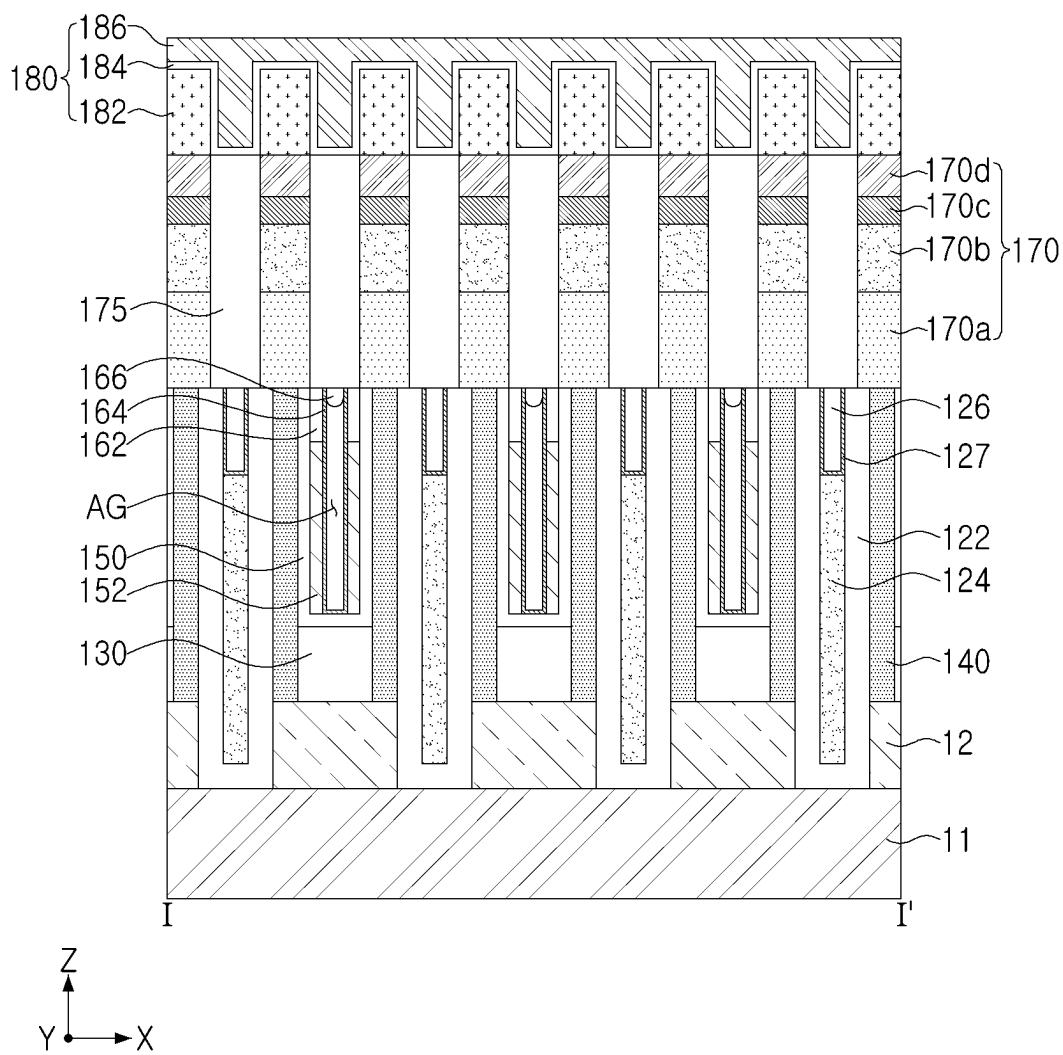


FIG. 15

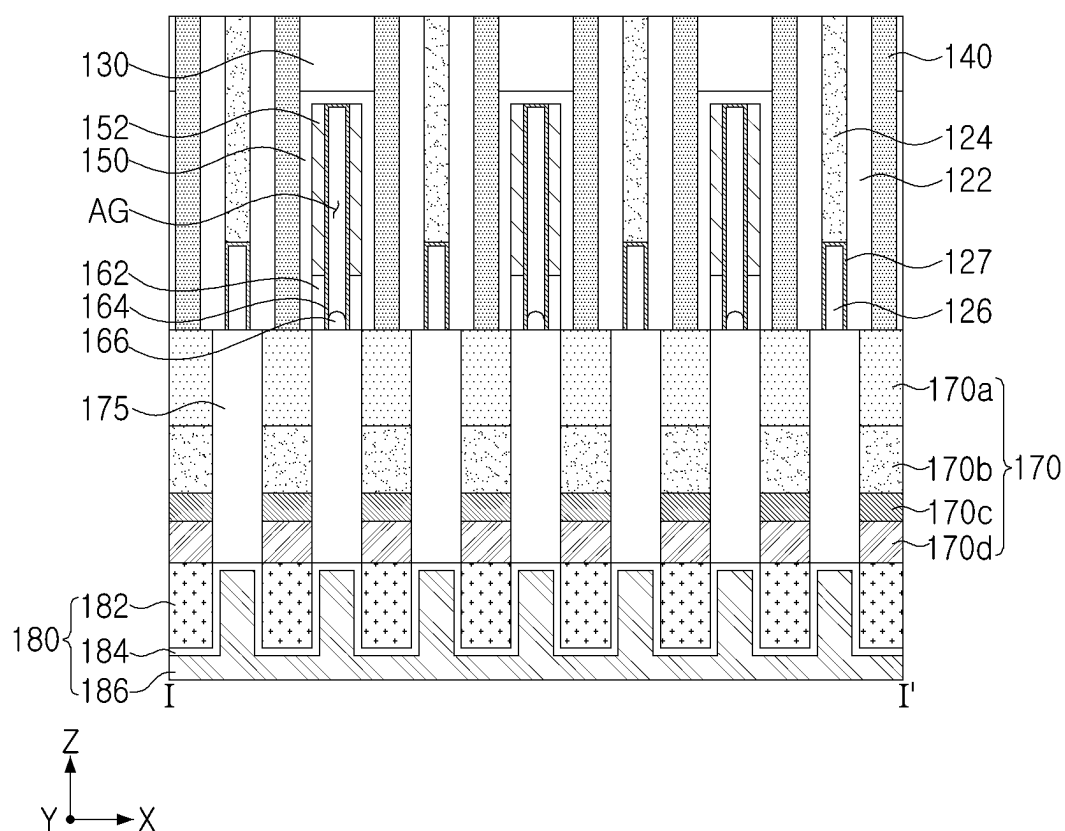


FIG. 16

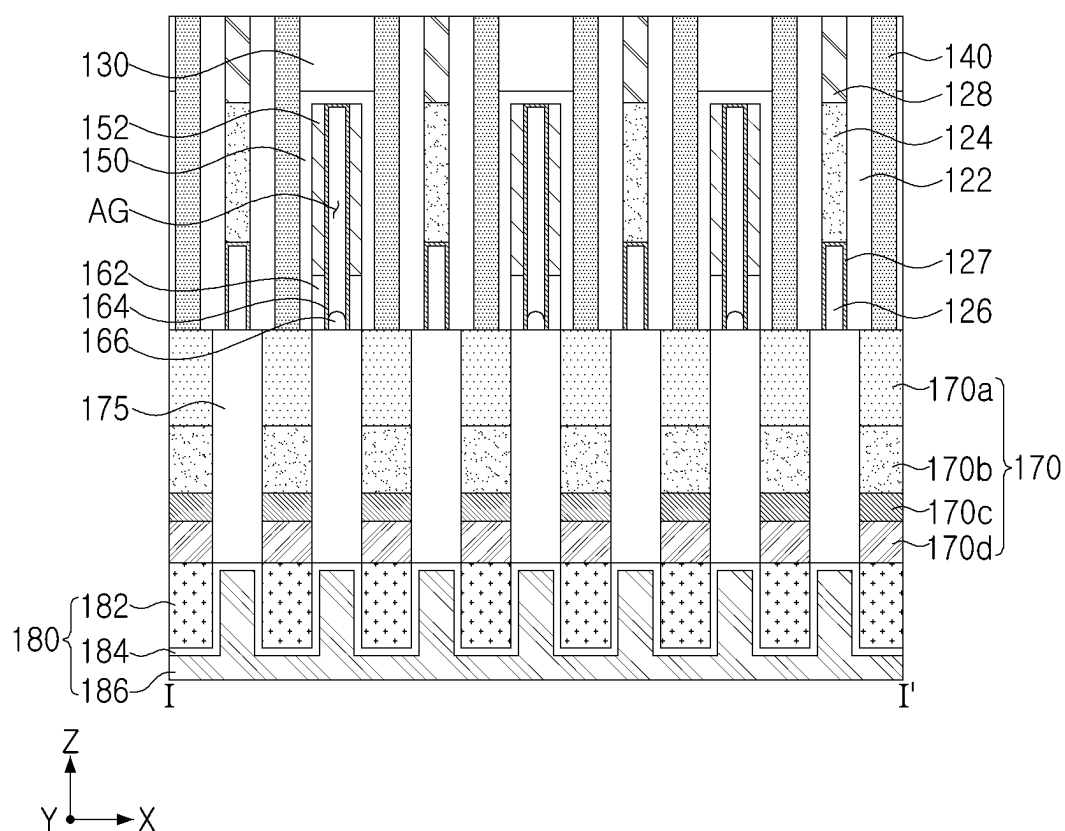


FIG. 17

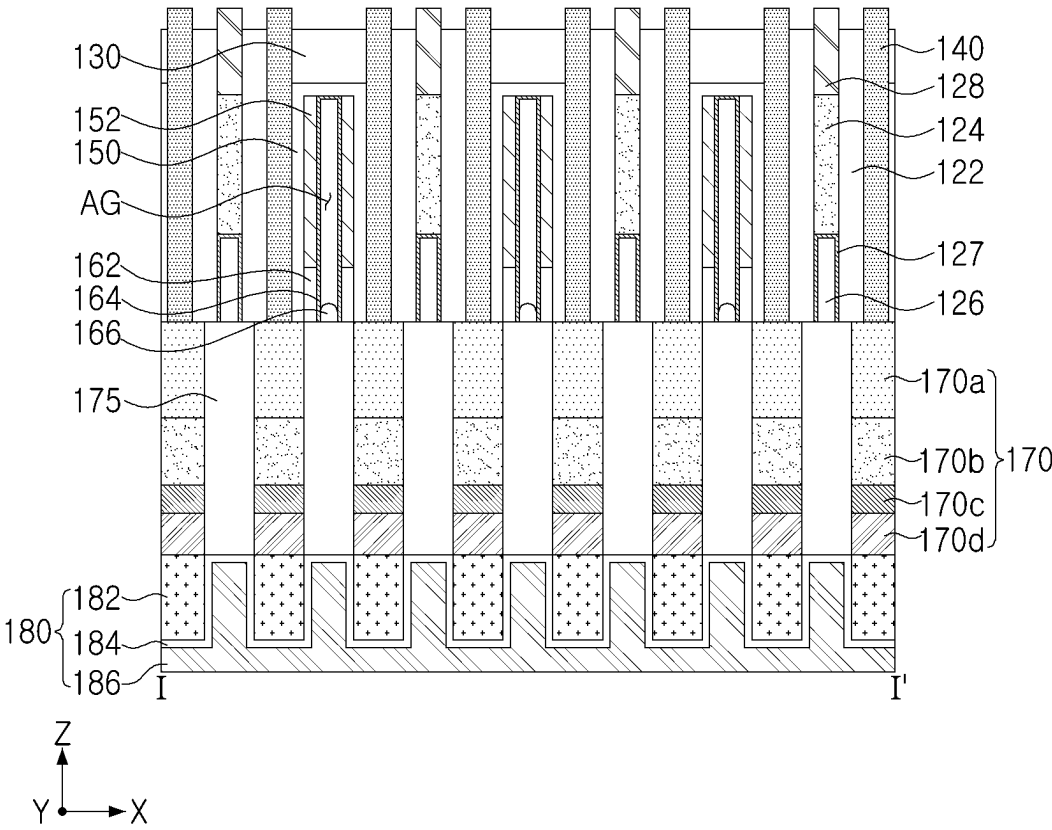


FIG. 18

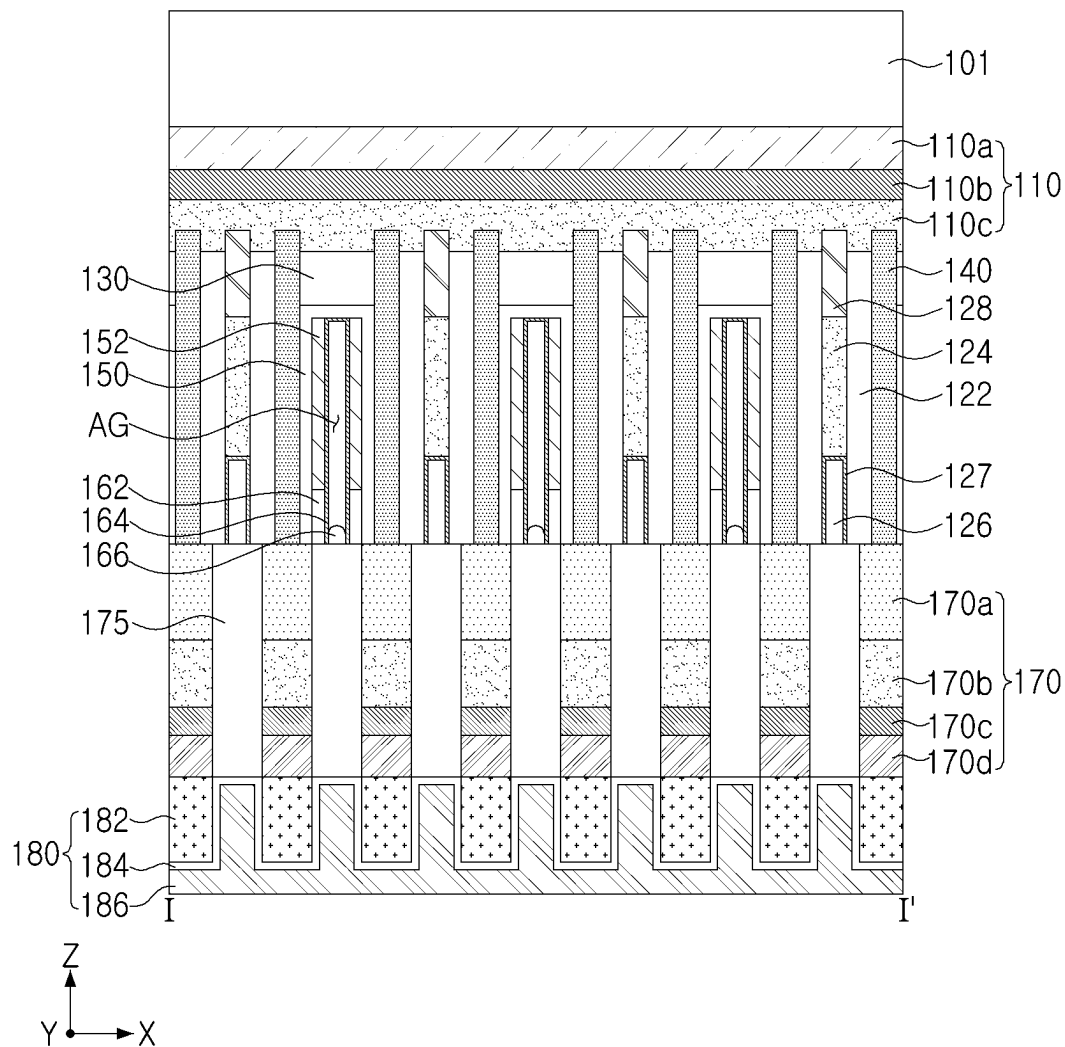


FIG. 19

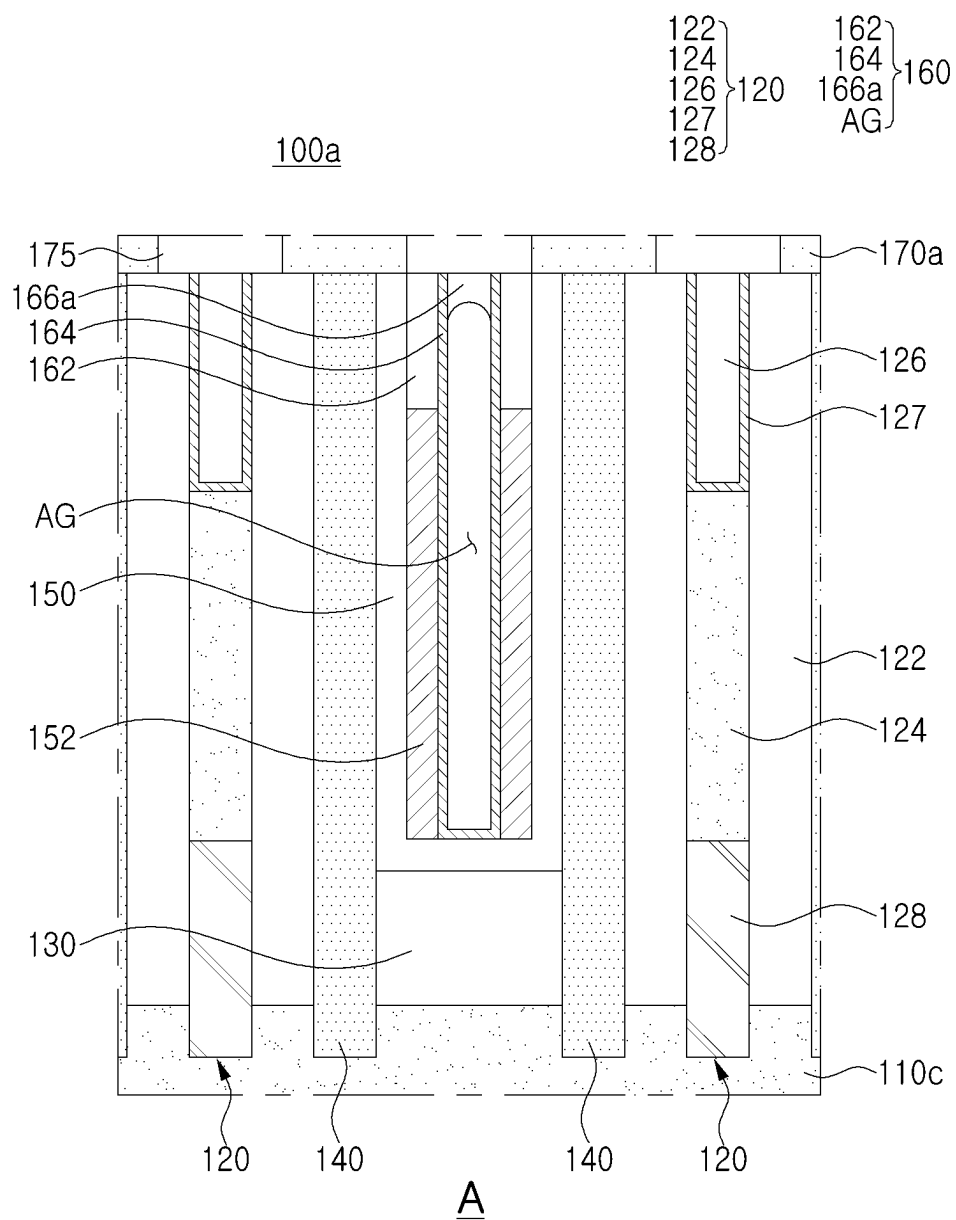


FIG. 20

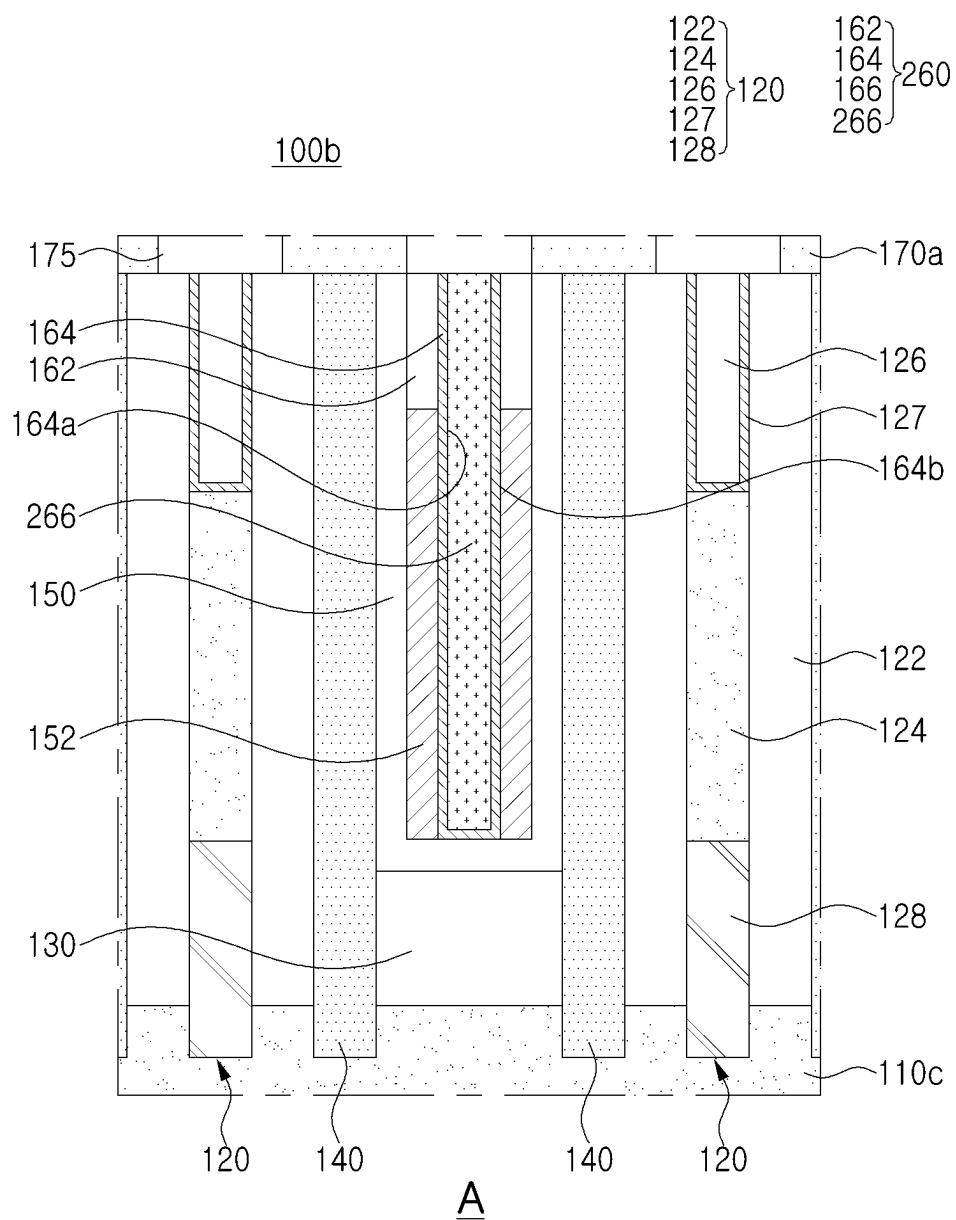


FIG. 21

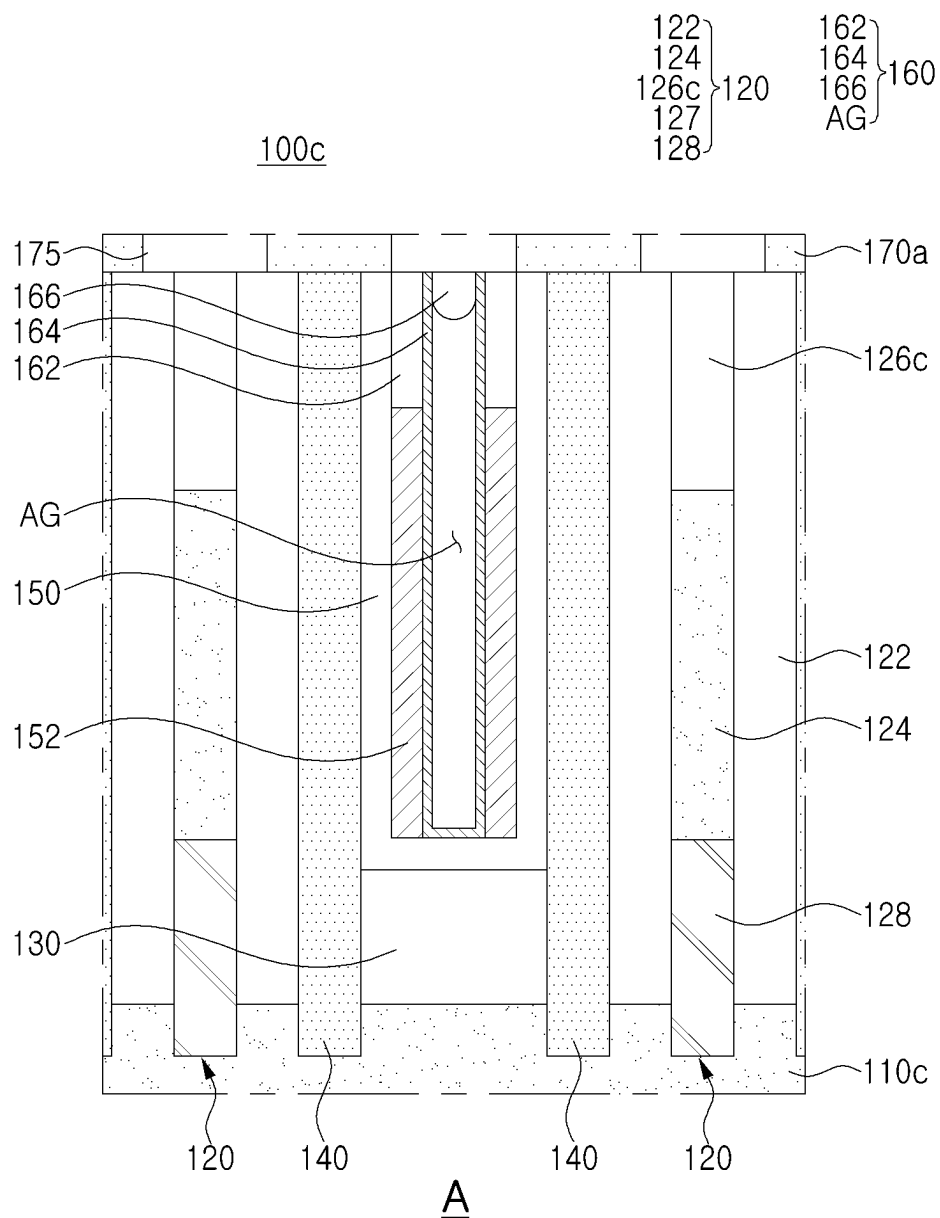


FIG. 22

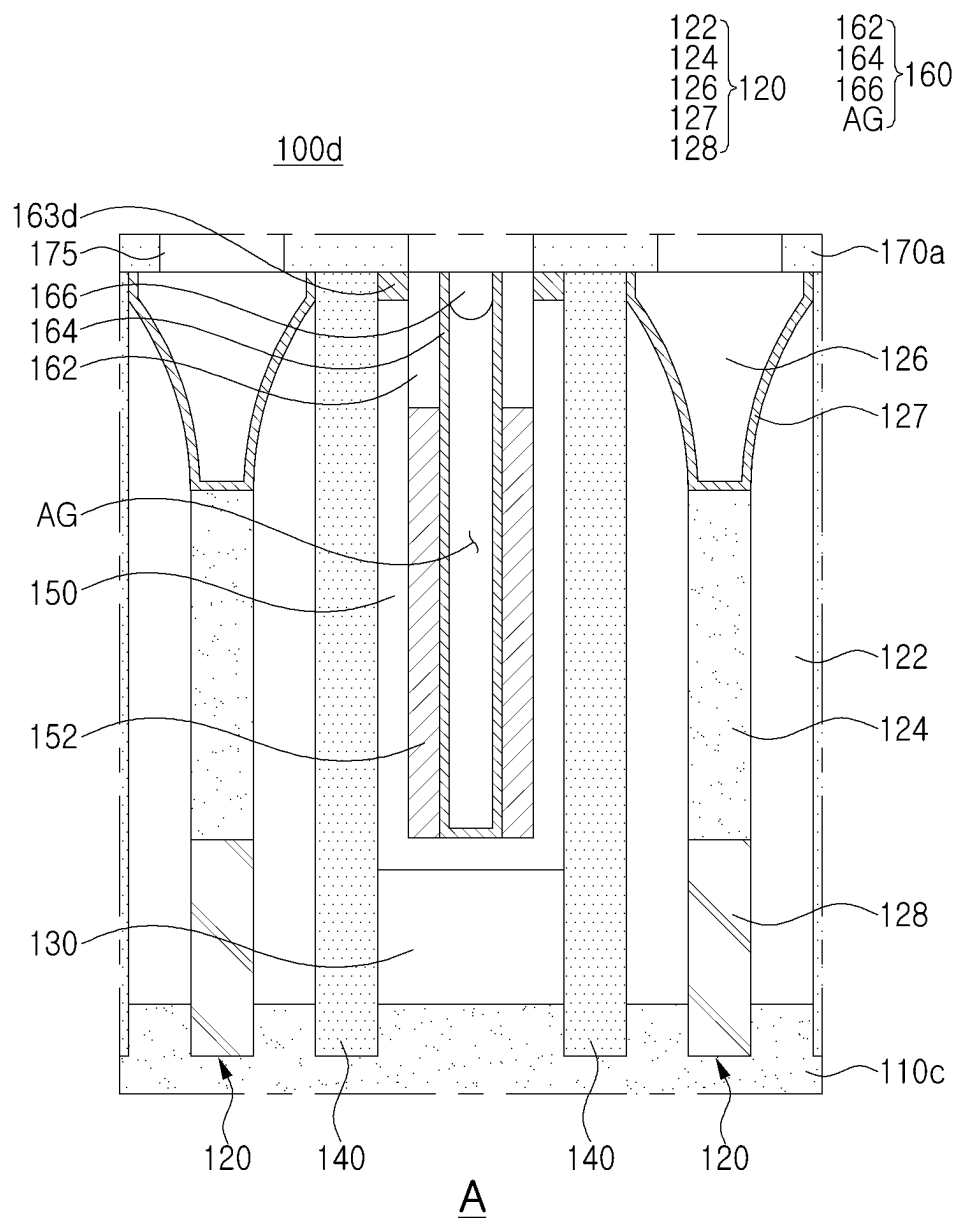


FIG. 23

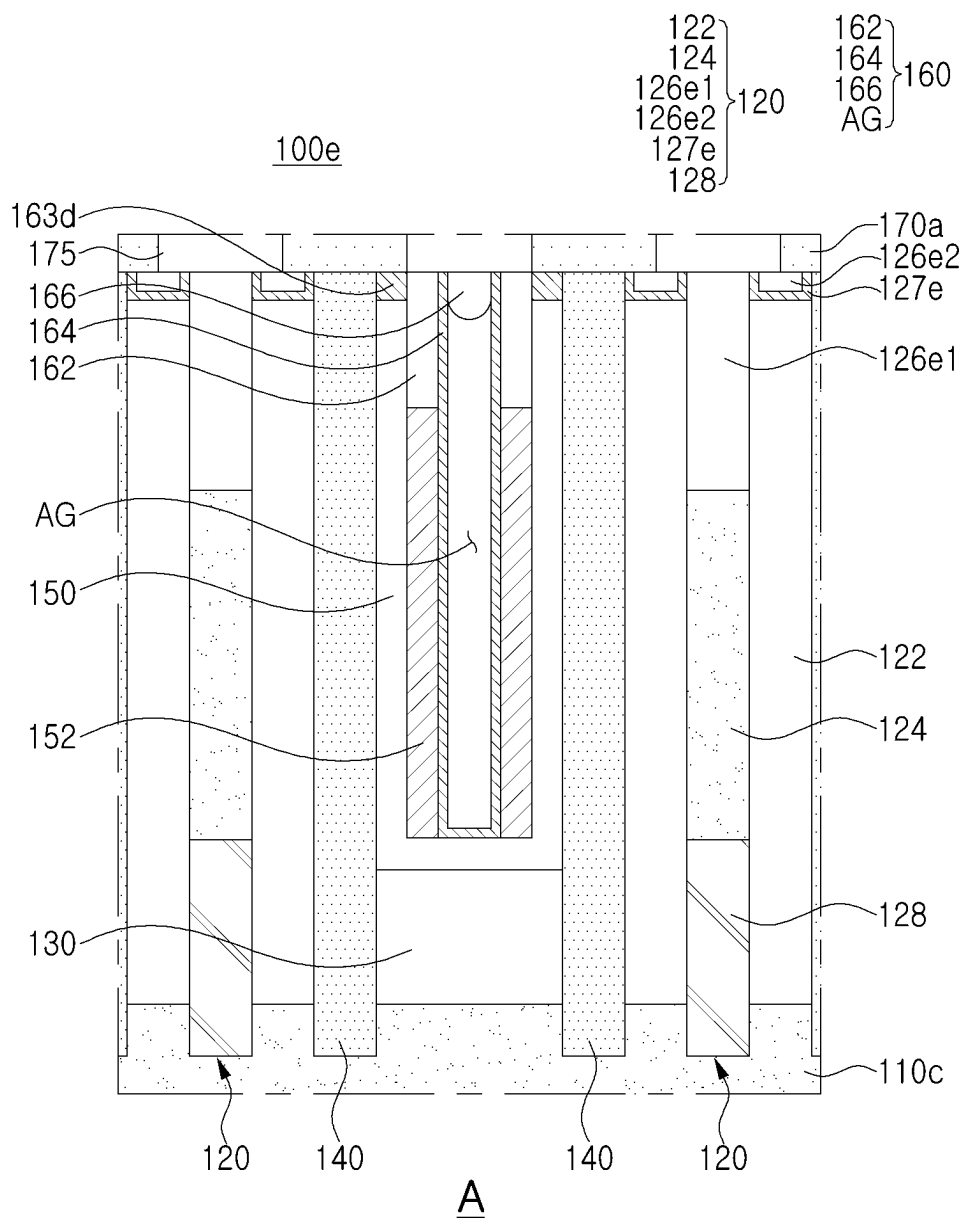


FIG. 24

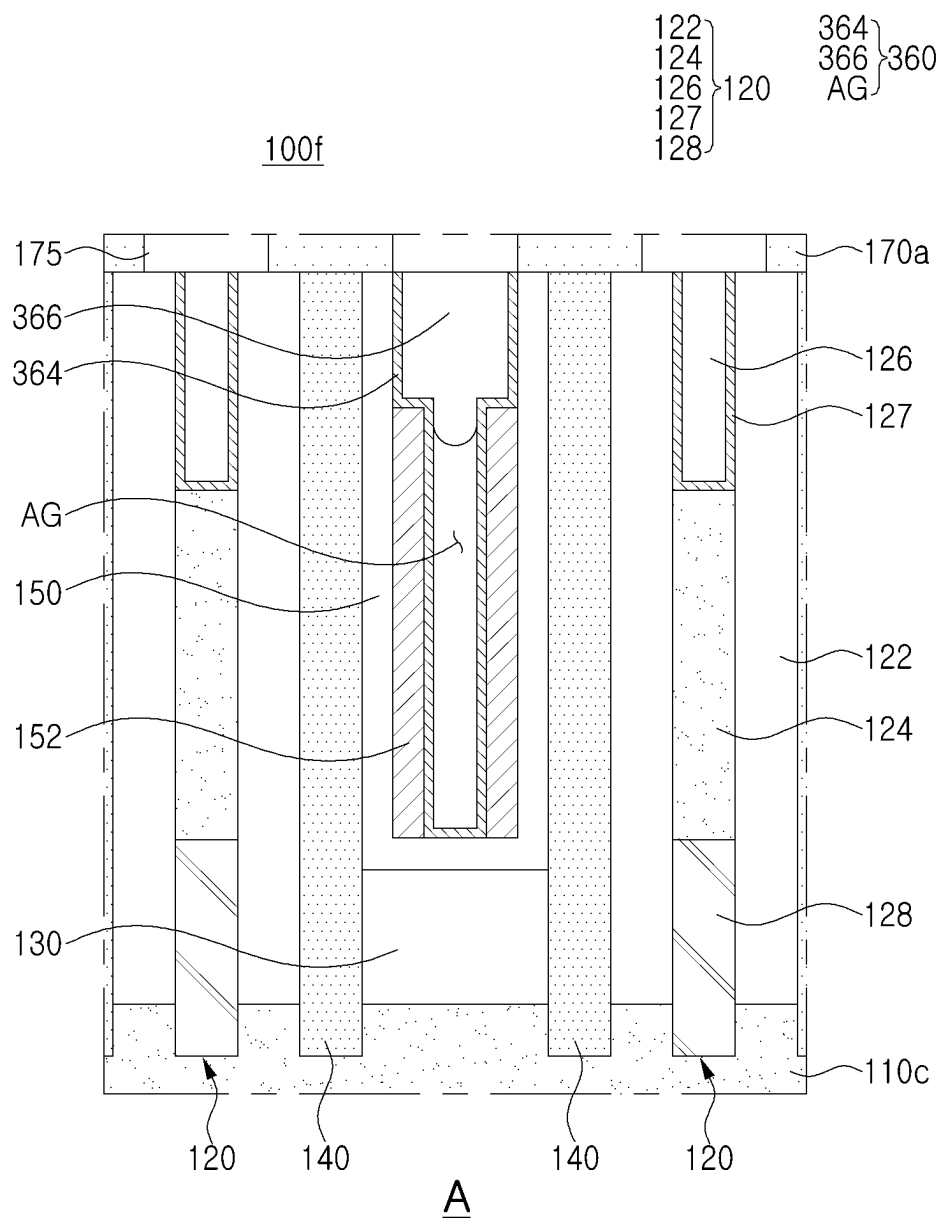


FIG. 25

SEMICONDUCTOR DEVICES HAVING INSULATING STRUCTURES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims benefit of priority to Korean Patent Application No. 10-2024-0032013, filed on Mar. 6, 2024 in the Korean Intellectual Property Office, the inventive concept of which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The present inventive concept relates to semiconductor devices having insulating structures.

[0003] As demand for high performance, high speed, and/or multifunctionality of semiconductor devices increases, a degree of integration of semiconductor devices is increasing. In manufacturing fine-patterned semiconductor devices in response to the trend for high integration of semiconductor devices, it may be required to implement patterns with fine widths or fine spacings.

SUMMARY OF THE INVENTION

[0004] An aspect of the technical problems to be solved by the technical concept of the present inventive concept is to provide a semiconductor device having an insulating structure including an air gap or a low- κ dielectric material disposed between word lines.

[0005] According to aspects of the present inventive concept, a semiconductor device, comprising: a bit line structure; back gate structures on the bit line structure; a first channel layer and a second channel layer between the back gate structures in a first direction, wherein the first channel layer and the second channel layer extend in a second direction; a first word line and a second word line between the first channel layer and the second channel layer in the first direction, wherein the first word line is adjacent to the first channel layer, and the second word line is adjacent to the second channel layer; and an insulating structure on a side surface of the first word line, a side surface of the second word line, an upper surface of the first word line, and an upper surface of the second word line, wherein the insulating structure includes a core region between the first word line and the second word line in the first direction; and an insulating liner between the core region and the first word line in the first direction and between the core region and the second word line in the first direction, wherein the side surface of the first word line and the side surface of the second word line face each other in the first direction, wherein the insulating liner extends to a level higher than those of the first word line and the second word line, wherein the first direction is parallel with a lower surface of the bit line structure, and wherein the second direction is perpendicular to the lower surface of the bit line structure.

[0006] According to aspects of the present inventive concept, a semiconductor device, comprising: a bit line structure; a channel layer on the bit line structure, wherein the channel layer extends in a first direction; a back gate structure facing a first side surface of the channel layer in a second direction; a word line facing a second side surface of the channel layer that is opposite to the first side surface of the channel layer in the second direction; a gate dielectric layer between the channel layer and the word line in the

second direction; and an insulating structure on the word line, wherein the gate dielectric layer includes a vertical portion in contact with a first side surface of the word line and a side surface of the insulating structure, and a horizontal portion in contact with a lower surface of the word line and a lower surface of the insulating structure, wherein the insulating structure includes a core region spaced apart from the word line; and an insulating liner between the core region and a second side surface of the word line in the second direction, and extending between the horizontal portion and the core region in the first direction, wherein the first side surface of the word line is opposite to the second side surface of the word line in the second direction, wherein the first direction is perpendicular to a lower surface of the bit line structure, and wherein the second direction is parallel with the lower surface of the bit line structure.

[0007] According to aspects of the present inventive concept, a semiconductor device, comprising: a bit line structure; back gate structures on the bit line structure; a first channel layer and a second channel layer between the back gate structures in a first direction, wherein the first channel layer and the second channel layer extend in a second direction; a first word line and a second word line between the first channel layer and the second channel layer in the first direction, wherein the first word line is adjacent to the first channel layer, and the second word line is adjacent to the second channel layer; an insulating structure in contact with a side surface of the first word line, a side surface of the second word line, an upper surface of the first word line, and an upper surface of the second word line; a gate dielectric layer between the first channel layer and the first word line in the first direction, between the first channel layer and a first upper region of the insulating structure in the first direction, between the second channel layer and the second word line in the first direction, and between the second channel layer and a second upper region of the insulating structure in the first direction; contact patterns electrically connected to the first channel layer and the second channel layer; and an information storage structure on the contact patterns, wherein the insulating structure includes a core region spaced apart from the first word line and second word line; and an insulating liner between the side surface of the first word line and the core region and between the side surface of the second word line and the core region, and extending to a level higher than those of the first word line and the second word line, wherein the core region has a dielectric constant lower than that of silicon oxide, wherein the first upper region of the insulating structure is on the first word line, wherein the second upper region of the insulating structure is on the second word line, wherein the side surface of the first word line and the side surface of the second word line face each other in the first direction, wherein the first direction is parallel with a lower surface of the bit line structure, and wherein the second direction is perpendicular to the lower surface of the bit line structure.

BRIEF DESCRIPTION OF DRAWINGS

[0008] The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings:

[0009] FIG. 1 is a plan view of a semiconductor device according to an example embodiment;

[0010] FIG. 2 is a cross-sectional view taken along line I-I' of the semiconductor device shown in FIG. 1;

[0011] FIG. 3 is an enlarged view of a portion of the semiconductor device shown in FIG. 2;

[0012] FIGS. 4 to 19 are cross-sectional views shown according to a process sequence to illustrate a method of manufacturing a semiconductor device according to an example embodiment; and

[0013] FIGS. 20 to 25 are cross-sectional views of semiconductor devices according to example embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0014] Hereinafter, example embodiments of the present inventive concept will be described with reference to the attached drawings.

[0015] FIG. 1 is a plan view of a semiconductor device according to an example embodiment. FIG. 2 is a cross-sectional view taken along line I-I' of the semiconductor device shown in FIG. 1. FIG. 3 is an enlarged view of a portion of the semiconductor device shown in FIG. 2. FIG. 3 may correspond to region A of FIG. 2.

[0016] Referring to FIGS. 1 to 3, a semiconductor device 100 according to an example embodiment of the present disclosure may include a lower insulating layer 101, a bit line structure 110, a back gate structure 120, a channel layer 140, a word line 152, an insulating structure 160, a contact pattern 170, and an information storage structure 180.

[0017] The semiconductor device 100 may include a channel layer 140, a bit line structure 110 electrically connected to the channel layer 140, and a vertical channel transistor comprised of word lines 152 disposed on at least one side of the channel layer 140. It will be understood that when an element or layer is referred to as being “on”, “responsive to”, “connected to”, or “coupled to” another element or layer, it may be directly on, responsive to, connected to, or coupled to the other element or layer, or one or more intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on”, “directly responsive to”, “directly connected to”, or “directly coupled to”, another element, there are no intervening elements present. In addition, “electrical connection” conceptually includes a direct physical connection and an indirect physical connection.

[0018] The semiconductor device 100 may be applied to, for example, a cell array of Dynamic Random Access Memory (DRAM), but the present inventive concept is not limited thereto.

[0019] The lower insulating layer 101 may include an insulating material such as silicon oxide, silicon nitride, silicon oxynitride (SiON), silicon carbonitride (SiCN), or the like.

[0020] The bit line structure 110 may extend in an X-direction on the lower insulating layer 101. In an example embodiment, the bit line structure 110 may be buried within the lower insulating layer 101. The bit line structure 110 may be electrically connected to the channel layer 140.

[0021] A plurality of bit line structures 110 may be provided, and the plurality of bit line structures 110 may be spaced apart from each other in a Y-direction and extend in parallel. The X-direction and the Y-direction may be parallel with an upper surface of the lower insulating layer 101 (or a lower surface of the bit line structure 110). The X-direction and the Y-direction may intersect each other. For example,

the X-direction and the Y-direction may be perpendicular to each other. A Z-direction may be perpendicular to the upper surface of the lower insulating layer 101 (or the lower surface of the bit line structure 110).

[0022] The bit line structure 110 may include, for example, doped polysilicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, conductive graphene, carbon nanotube, and/or a combination thereof. For example, the bit line structure 110 may include doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, graphene, carbon nanotube, and/or a combination thereof. In an example embodiment, the bit line structure 110 may include a first conductive pattern 110a, a second conductive pattern 110b, and a third conductive pattern 110c, sequentially stacked on the lower insulating layer 101. The first conductive pattern 110a may include, for example, a metal material such as titanium (Ti), tantalum (Ta), tungsten (W), and aluminum (Al), the second conductive pattern 110b may include, for example, a metal nitride such as titanium nitride (TiN) or a silicide material such as titanium silicide (TiSi), and the third conductive pattern 110c may include, for example, a semiconductor material such as polycrystalline silicon. The third conductive pattern 110c may be a layer doped with impurities. However, depending on example embodiments, materials, the number of layers, and thicknesses of the layers forming the bit line structure 110 may vary. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0023] The back gate structures 120 may intersect (overlap) the bit line structures 110. For example, the back gate structures 120 may extend in the Y-direction, and may be spaced apart from each other in the X-direction.

[0024] The back gate structure 120 may include a back gate dielectric layer 122, a back gate electrode 124, an upper capping layer 126, and a lower capping layer 128. The back gate electrodes 124 may extend in the Y-direction and may be spaced apart from each other in the X-direction. The back gate electrode 124 may serve to remove charges trapped in the channel layer 140. The channel layer 140 may be a floating body, and the back gate electrode 124 may be a structure to supplement the floated channel layer 140 to prevent or minimize performance degradation of the semiconductor device 100 due to a floating body effect of the channel layer 140.

[0025] The back gate electrode 124 may include, for example, doped polysilicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, graphene, carbon nanotube, and/or a combination thereof. For example, the back gate electrode 124 may comprise doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiAlC, TaAlC, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, graphene, carbon nanotube, and/or a combination thereof, but the present inventive concept is not limited thereto. The back gate electrode 124 may be formed of a single layer or multiple layers of the above-described materials.

[0026] In an example embodiment, the back gate electrode 124 may include (e.g., may be formed of) the same material as the word line 152, but is not limited thereto and may include other materials.

[0027] The back gate dielectric layers 122 may extend in the Y-direction along both side surfaces (e.g., opposite side

surface in the X-direction) of the back gate electrodes **124**. A (vertical) length of the back gate dielectric layer **122** (in the Z-direction) may be greater than a (vertical) length of the back gate electrode **124** (in the Z-direction). For example, an upper surface of the back gate dielectric layer **122** may be located on a level higher than that of an upper surface of the back gate electrode **124**, and a lower surface of the back gate dielectric layer **122** may be located on a level higher than that of a lower surface of the back gate electrode **124**. The lower surface of the back gate dielectric layer **122** may be in contact with the third conductive pattern **110c**. Each of the back gate dielectric layers **122** may include, for example, silicon oxide and/or high- κ dielectric. A high- κ dielectric (material) may refer to a material having a higher dielectric constant than that of silicon oxide. Herein, the level may be a relative location (e.g., distance) from a lower surface of the lower insulating layer **101** (or a lower surface of the bit line structure **110**) in a vertical direction (e.g., in the Z-direction). A farther distance from the lower surface of the lower insulating layer **101** (or from the lower surface of the bit line structure **110**) may be a higher level. A closer distance from the lower surface of the lower insulating layer **101** (or from the lower surface of the bit line structure **110**) may be a lower level.

[0028] The upper capping layer **126** may be disposed on the back gate electrode **124**. An upper surface of the upper capping layer **126** may be coplanar with an upper surface of the back gate dielectric layer **122**. In an example embodiment, the back gate structure **120** may further include a back gate liner **127** on (e.g., covering or overlapping) a lower surface and side surfaces of the upper capping layer **126**. The back gate liner **127** may be formed on (conformally along) a side surface of the back gate dielectric layer **122** and an upper surface of the back gate electrode **124**. The upper capping layer **126** and the back gate liner **127** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, a low- κ dielectric material, and/or a combination thereof. For example, the upper capping layer **126** and the back gate liner **127** may include silicon nitride. A low- κ dielectric (material) may refer to a material having a lower dielectric constant than that of silicon oxide.

[0029] The lower capping layer **128** may be disposed below the back gate electrode **124**. For example, the lower capping layer **128** may be on a lower surface of the back gate electrode **124**. A lower surface of the lower capping layer **128** may be located on a level lower than that of the lower surface of the back gate dielectric layer **122**.

[0030] The channel layer **140** may be disposed on (in) the bit line structure **110**, and may extend in the vertical direction (e.g., Z-direction). In some embodiments, the channel layer **140** may extend in at least a portion of the bit line structure **110** (e.g., the third conductive pattern **110c**). In a plan view, the channel layers **140** may be disposed on both sides (e.g., opposite sides in the X-direction) of the back gate structures **120**. The channel layers **140** may be spaced apart from each other in the X- and Y-directions (in a plan view). An upper surface of the channel layer **140** may be coplanar with the upper surface of the back gate structure **120**. A lower surface of the channel layer **140** may be in contact with the third conductive pattern **110c** and may be located on a level lower than that of the lower surface of the back gate dielectric layer **122**. In some embodiments, the lower surface of the back gate dielectric layer **122** may be in contact

with the upper surface of the bit line structure **110** (e.g., the upper surface of the third conductive pattern **110c**).

[0031] Each of the channel layers **140** may include a first source/drain region in contact with the bit line structure **110** and a second source/drain region connected (e.g., electrically connected) to the contact pattern **170**. In an example embodiment, the first and second source/drain regions may have an N-type conductivity type.

[0032] In an example embodiment, the channel layers **140** may include a monocrystalline semiconductor material. The monocrystalline semiconductor material may include a group IV semiconductor, a group III-V compound semiconductor, and/or a group II-VI compound semiconductor, and may be, for example, a single crystal including silicon, silicon carbide, germanium, and/or silicon-germanium.

[0033] However, depending on example embodiments, the channel layers **140** may include a polycrystalline semiconductor material layer, an oxide semiconductor material layer such as Indium Gallium Zinc Oxide (IGZO), and/or a two-dimensional material layer such as MoS₂, or the like.

[0034] The oxide semiconductor material layer may include indium gallium zinc oxide (IGZO). However, the example embodiment thereof is not limited thereto. For example, the oxide semiconductor material layer may include Indium Tungsten Oxide (IWO), Indium Tin Gallium Oxide (ITGO), Indium Aluminum Zinc Oxide (IAZO), Indium Gallium Oxide (IGO), Indium Tin Zinc Oxide (ITZO), zinc tin oxide (ZTO), indium zinc oxide (IZO), ZnO, indium gallium silicon oxide (IGSO), indium oxide (InO), tin oxide (SnO), titanium oxide (TiO), zinc oxynitride (ZnON), magnesium zinc oxide (MgZnO), indium zinc oxide (InZnO), indium gallium zinc oxide (InGaZnO), zirconium indium zinc oxide (ZrInZnO), hafnium indium zinc oxide (HfInZnO), tin indium zinc oxide (SnInZnO), aluminum tin indium zinc oxide (AlSnInZnO), silicon indium zinc oxide (SiInZnO), zinc tin oxide (ZnSnO), aluminum zinc tin oxide (AlZnSnO), gallium zinc tin oxide (GaZnSnO), zirconium zinc tin oxide (ZrZnSnO), and/or indium gallium silicon oxide (InGaSiO).

[0035] The two-dimensional material layer may include, for example, a TMD material layer (Transition Metal Dichalcogenide material layer), a black phosphorous material layer, and/or a hBN material layer (hexagonal Boron-Nitride material layer) which may have semiconductor properties. For example, the two-dimensional material layer may include BiOSe, CrI, WSe₂, MoS₂, TaS, WS, SnSe, ReS, β -SnTe, MnO, AsS, P(black), InSe, h-BN, GaSe, GaN, SrTiO, MXene, and/or Janus 2D materials which may form two-dimensional materials.

[0036] The word lines **152** may be disposed on the bit line structure **110**, and may be disposed on both sides (e.g., opposite sides in the X-direction) of the back gate structure **120**. The word lines **152** may be spaced apart from each other in the X-direction. In a plan view, the word line **152** may extend around (e.g., surround) at least a portion of channel layers **140**, and the channel layers **140** may be disposed between the back gate structure **120** and the word line **152**. The word line **152** may include a first word line **152_1** and a second word line **152_2** disposed between two adjacent back gate structures **120** (in the X-direction). The channel layer **140** may include a first channel layer **140_1** and a second channel layer **140_2** disposed between the two adjacent back gate structures **120** (in the X-direction), and the first word line **152_1** and the second word line **152_2**

may be disposed between the first channel layer **140_1** and the second channel layer **140_2** (in the X-direction). The first word line **152_1** may be adjacent the first channel layer **140_1**, and the second word line **152_2** may be adjacent the second channel layer **140_2**.

[0037] The word line **152** may include, for example, doped polysilicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, conductive graphene, carbon nanotube, and/or a combination thereof. For example, the word line **152** may include doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, graphene, carbon nanotubes, and/or a combination thereof, but is not limited thereto. The word line **152** may include a single layer or multiple layers of the materials described above.

[0038] The semiconductor device **100** may further include a gate dielectric layer **150** and an insulating structure **160**. The gate dielectric layer **150** may be disposed between the word line **152** and the channel layer **140** (in the X-direction) and may have a U-shape or a similar shape in a cross-sectional view. For example, the gate dielectric layer **150** may be disposed below (on) the lower surface of the insulating structure **160** and may (include portions that) extend between the first channel layer **140_1** and the first word line **152_1** and between the second channel layer **140_2** and the two word lines **152_2** (in the X-direction and/or the Y-direction). In some embodiments, in a plan view, the gate dielectric layer **150** may include portions that extend between the back gate structure **120** and the word line **152** (the first word line **152_1** or the second word line **152_2**) without the channel layer **140** (the first channel layer **140_1** or the second channel layer **140_2**) therebetween.

[0039] The gate dielectric layer **150** may include a first gate dielectric layer **150_1** and a second gate dielectric layer **150_2**. The first gate dielectric layer **150_1** may include a first vertical portion **150_1a** and a first horizontal portion **150_1b**. The first vertical portion **150_1a** may include at least a portion that extends in the Z-direction between the first channel layer **140_1** and the first word line **152_1**. In some embodiments, the first vertical portion **150_1a** may include a portion that extends in the Z-direction between the first channel layer **140_1** and a first gate capping layer **162a** (to be described later) on the first word line **152_1**. For example, the first vertical portion **150_1a** may be in contact with a side surface of the first channel layer **140_1**, a side surface of the first word line **152_1**, and/or a side surface of the first gate capping layer **162a**. In some embodiments, an upper surface of the first vertical portion **150_1a** may be a portion of an upper (e.g., the uppermost) surface of the gate dielectric layer **150** (e.g., a portion of an upper (e.g., the uppermost) surface of the first gate dielectric layer **150_1**). A lower surface of the first vertical portion **150_1a** may be a portion of a lower surface of the gate dielectric layer **150** (e.g., a portion of a lower surface of the first gate dielectric layer **150_1**) or in contact with the first horizontal portion **150_1b**. The first horizontal portion **150_1b** may extend in the X-direction between the first channel layer **140_1** and a second horizontal portion **150_2b** (to be described later) or between the first vertical portion **150_1a** and the second horizontal portion **150_2b**. The first horizontal portion **150_1b** may include a portion that is disposed between the first word line **152_1** and the bit line structure **110** (or a lower insulating pattern **130** (to be described later)) in the Z-direction.

A lower surface of the first horizontal portion **150_1b** may be a portion of the lower surface of the gate dielectric layer **150** (e.g., a portion of the lower surface of the first gate dielectric layer **150_1**). In some embodiments, the first horizontal portion **150_1b** may be in contact with the lower surface of the first word line **152_1** and the lower surface of the insulating structure **160** (e.g., a lower surface of an insulating liner **164**, which will be described later). The second gate dielectric layer **150_2** may include a second vertical portion **150_2a** and a second horizontal portion **150_2b**. The second vertical portion **150_2a** may include at least a portion that extends in the Z-direction between the second channel layer **140_2** and the second word line **152_2**. In some embodiments, the second vertical portion **150_2a** may include a portion that extends in the Z-direction between the second channel layer **140_2** and a second gate capping layer **162b** (to be described later) on the second word line **152_2**. For example, the second vertical portion **150_2a** may be in contact with a side surface of the second channel layer **140_2**, a side surface of the second word line **152_2**, and/or a side surface of the second gate capping layer **162b**. In some embodiments, an upper surface of the second vertical portion **150_2a** may be a portion of an upper (e.g., the uppermost) surface of the gate dielectric layer **150** (e.g., a portion of an upper (e.g., the uppermost) surface of the second gate dielectric layer **150_2**). A lower surface of the second vertical portion **150_2a** may be a portion of a lower surface of the gate dielectric layer **150** (e.g., a portion of a lower surface of the second gate dielectric layer **150_2**) or in contact with the second horizontal portion **150_2b**. The second horizontal portion **150_2b** may extend in the X-direction between the second channel layer **140_2** and the first horizontal portion **150_1b** or between the second vertical portion **150_2a** and the first horizontal portion **150_1b**. The second horizontal portion **150_2b** may include a portion that is disposed between the second word line **152_2** and the bit line structure **110** (or the lower insulating pattern **130**) in the Z-direction. A lower surface of the second horizontal portion **150_2b** may be a portion of the lower surface of the gate dielectric layer **150** (e.g., a portion of the lower surface of the second gate dielectric layer **150_2**). In some embodiments, the second horizontal portion **150_2b** may be in contact with the lower surface of the second word line **152_2** and the lower surface of the insulating structure **160** (e.g., the lower surface of the insulating liner **164**, which will be described later). In some embodiments, an upper surface (e.g., the uppermost surface) of the gate dielectric layer **150** may be coplanar with an upper surface of the channel layer **140**. The first vertical portion **150_1a**, the first horizontal portion **150_1b**, the second vertical portion **150_2a**, and the second horizontal portion **150_2b** may form an integrated unitary structure (the gate dielectric layer **150**) having no visible boundaries therebetween.

[0040] In an example, each of the gate dielectric layers **150** may be a tunnel dielectric layer not including an information storage layer. For example, each of the gate dielectric layers **150** may include, for example, silicon oxide and/or a high- κ dielectric material. The high- κ dielectric material may include metal oxide and/or metal oxynitride. For example, the high- κ dielectric material may include HfO_2 , HfSiO , HfSiON , HfTaO , HfTiO , HfZrO , ZrO_2 , Al_2O_3 , and/or a combination thereof, but the present disclosure

sure is not limited thereto. Each of the gate dielectric layers **150** may be formed of a single layer or multiple layers of the above-described materials.

[0041] In another example, each of the gate dielectric layers **150** may include an information storage layer and a dielectric layer. For example, each of the gate dielectric layers **150** may have polarization characteristics depending on an electric field and may include a ferroelectric layer that may have remnant polarization due to a dipole even in the absence of an external electric field. Data may be recorded using the polarization state within the ferroelectric layer. Accordingly, each of the gate dielectric layers **150** may include a ferroelectric layer, which may be referred to as an information storage layer. The ferroelectric layer, which may be the information storage layer, may include, for example, an Hf-based compound, a Zr-based compound, and/or an Hf-Zr-based compound. For example, the Hf-based compound may include a HfO-based ferroelectric material, the Zr-based compound may include a ZrO-based ferroelectric material, and the Hf-Zr-based compound may include a hafnium zirconium oxide (HZO)-based ferroelectric material. The ferroelectric layer, which may be the information storage layer, may include a ferroelectric material doped with impurities, for example, C, Si, Mg, Al, Y, N, Ge, Sn, Gd, La, Sc, and/or Sr. For example, the ferroelectric layer, which may be the information storage layer, may include a material in which HfO₂, ZrO₂, and/or HfZrO is doped with impurities such as C, Si, Mg, Al, Y, N, Ge, Sn, Gd, La, Sc, and/or Sr.

[0042] In the gate dielectric layers **150**, the information storage layer is not limited to the types of the above-described materials, and may include a material that can store information.

[0043] In a cross-sectional view, the insulating structure **160** may be disposed between the (adjacent) back gate structures **120** (in the X-direction). For example, (each of) the insulating structures **160** may include at least a portion that extends in the Y-direction between adjacent word lines **152** (in the X-direction) and may be spaced apart from each other in the X-direction. For example, at least a portion of the insulating structure **160** may extend in the Y-direction between the first word line **152_1** and the second word line **152_2** in the X-direction. The insulating structure **160** may be disposed on the word lines **152** and may extend between the word lines **152**. For example, the insulating structure **160** may contact side surfaces of the first and second word lines **152_1** and **152_2** facing each other (in the X-direction) and upper surfaces of the first and second word lines **152_1** and **152_2**.

[0044] In an example embodiment, the insulating structure **160** may include a core region, a gate capping layer **162**, an insulating liner **164**, and a capping pattern **166**. The core region may include a region disposed between the first and second word lines **152_1** and **152_2** (in the X-direction). The gate capping layers **162** may overlap the word lines **152** in a vertical direction (e.g., in the Z-direction), and may contact the gate dielectric layers **150**. The gate capping layers **162** may include a first gate capping layer **162a** on (e.g., overlapping in the Z-direction) the first word line **152_1** and a second gate capping layer **162b** on (e.g., overlapping in the Z-direction) the second word line **152_2**. In some embodiments, the core region may include a region disposed between the first gate capping layer **162a** and the second gate capping layer **162b**.

[0045] The insulating liner **164** may extend between the gate capping layers **162** (e.g., between the first gate capping layer **162a** and the second gate capping layer **162b**) and between the word lines **152** (e.g., between the first word line **152_1** and the second word line **152_2**). The insulating liner **164** may be disposed between the core area and the first word line **152_1**, and may extend to a level higher than that of the first word line **152_1**. The insulating liner **164** may be also disposed between the core region and the second word line **152_2**, and may extend to a level higher than that of the second word line **152_2**. For example, the insulating liner **164** may have a U-shape in a cross-sectional view, and may contact side surfaces of the gate capping layers **162** (e.g., the first gate capping layer **162a** and the second gate capping layer **162b**), side surfaces of the word lines **152** (e.g., the first word line **152_1** and the second word line **152_2**), and the gate dielectric layers **150** (e.g., the first horizontal portion **150_1b** and the second horizontal portion **150_2b**). The insulating liner **164** may extend from between the core region and the first word line **152_1** to between the core region and the first gate capping layer **162a**. The insulating liner **164** may also extend from between the core region and the second word line **152_2** to between the core region and the second gate capping layer **162b**. For example, the insulating liner **164** may extend in the Z-direction between the core region and the first gate capping layer **162a** and between the core region and the first word line **152_1**. The insulating liner **164** may extend in the Z-direction between the core region and the second gate capping layer **162b** and between the core region and the second word line **152_2**. The insulating liner **164** may extend from between the core region and the first word line **152_1** and from between the core region and the second word line **152_2** to between the core region and the gate dielectric layer **150**. For example, the insulating liner **164** may extend between the first word line **152_1** and the second word line **152_2** in the X-direction between the core region and the gate dielectric layer **150** (e.g., the first horizontal portion **150_1b** and/or the second horizontal portion **150_2b**) in the Z-direction.

[0046] The capping pattern **166** may be disposed between the gate capping layers **162** (e.g., between the first gate capping layer **162a** and the second gate capping layer **162b** in the X-direction) on the core region, and may be in contact with the insulating liner **164**. For example, the capping pattern **166** may contact an upper region of an inner surface **164a** of the insulating liner **164**. An outer surface **164b** of the insulating liner **164** may be in contact with the word lines **152** (e.g., the first word line **152_1** and the second word line **152_2**) and the gate capping layers **162** (e.g., the first gate capping layer **162a** and the second gate capping layer **162b**). The insulating liner **164** may extend between the capping pattern **166** and the first gate capping layer **162a** and between the capping pattern **166** and the second gate capping layer **162b**. A lower surface of the insulating liner **164** may be in contact with the gate dielectric layer **150**. A thickness of the insulating liner **164** may be greater than 0 Å and less than or equal to 10 Å. In an example embodiment, a lower surface of the capping pattern **166** may be convex (toward the core region in the Z-direction). Herein an upper region of an element may refer to a portion of the region of the element disposed farther than the central portion of the element from the lower surface of the lower insulating layer **101** in the Z-direction. A lower region of an element may refer to a portion of the region of the element disposed closer

than the central portion of the element to the lower surface of the lower insulating layer **101** in the Z-direction.

[0047] The gate capping layer **162**, the insulating liner **164**, and the capping pattern **166** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, low- κ dielectric material, and/or a combination thereof. In some embodiments, the insulating liner **164** may include the same material as the back gate liner **127**, for example, silicon nitride.

[0048] In an example embodiment, the insulating structure **160** may further include an air gap (AG) therein. For example, the core region may include (may be) an air gap AG. The air gap AG may refer to an empty (substantially empty) space or a space including (filled with) air and/or gas. The air gap AG may be defined by the insulating liner **164** and the capping pattern **166**. For example, a lateral limit (in the X-direction) and a lower limit (in the Z-direction) of the air gap AG may be defined by the insulating liner **164**, and an upper limit (in the Z-direction) of the air gap AG may be defined by the capping pattern **166**. For example, the air gap AG may be surrounded by the insulating liner **164** and the capping pattern **166**. In some example embodiments, the insulating structure **160** may include a low- κ dielectric material having a lower dielectric constant than silicon oxide therein. For example, the core region may include the low- κ dielectric material.

[0049] Since the insulating structure **160** (e.g., the core region) includes an air gap AG or a low- κ dielectric material therein, a dielectric constant of the core region may be lower than a dielectric constant of silicon oxide, and capacitance between the adjacent first word line **152_1** and the second word line **152_2** may be reduced. Accordingly, when one of the first word line **152_1** and the second word line **152_2** operates, a change in voltages of another word line may be reduced or prevented. That is, coupling noise between the adjacent word lines **152** (e.g., the first word line **152_1** and the second word line **152_2**) and coupling noise between the adjacent channel layers **140** (e.g., the first channel layer **140_1** and the second channel layer **140_2**) may be reduced. Therefore, when the semiconductor device **100** operates, leakage current may be prevented from increasing in an unselected transistor adjacent to the selected transistor. In addition, when an air gap AG or low- κ dielectric material is included therein, the capacitance of the insulating structure **160** is reduced, so a distance between the first word line **152_1** and the second word line **152_2** can be reduced, so that the semiconductor device **100** can be implemented with smaller design rules.

[0050] The gate capping layer **162**, the insulating liner **164**, and the capping pattern **166** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, low- κ dielectric material, and/or a combination thereof. For example, the gate capping layer **162**, the insulating liner **164**, and the capping pattern **166** may include silicon nitride.

[0051] The semiconductor device **100** may further include lower insulating patterns **130** disposed between the channel layers **140** (in the X-direction) and below the word lines **152**. The lower insulating patterns **130** may contact an upper surface of the third conductive pattern **110c**, side surfaces of the channel layers **140**, and lower surfaces of the gate dielectric layers **150**. Lower surfaces of the lower insulating patterns **130** may be located on a level higher than that of lower surfaces of the channel layers **140**. The lower insu-

lating pattern **130** may be on the third conductive pattern **110c**, and the gate dielectric layer **150** may be on the lower insulating pattern **130**. The lower insulating patterns **130** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, low dielectric constant, and/or a combination thereof. For example, the lower insulating patterns **130** may include silicon oxide.

[0052] The contact patterns **170** may be disposed on the channel layers **140** and may be electrically connected to the channel layers **140**. The contact patterns **170** may electrically connect the channel layers **140** and the information storage structure **180**.

[0053] Lower surfaces of the contact patterns **170** are shown as being in contact with the channel layer **140** and the gate dielectric layer **150**. However, according to example embodiments, the lower surfaces of the contact patterns **170** may also be in contact with the insulating structure **160** and/or the upper capping layer **126**. The lower surface of the contact patterns **170** may be in contact with the back gate structure **120**. In some embodiments, the contact patterns **170** may overlap with the back gate structure **120**, the channel layer **140**, the gate dielectric layer **150**, and/or the insulating structure **160** in the Z-direction.

[0054] The contact patterns **170** may include a conductive material, for example, doped monocrystalline silicon, doped polycrystalline silicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, conductive graphene, carbon nanotubes, and/or a combination thereof. In an example embodiment, the contact patterns **170** may include first, second, third, and fourth contact layers **170a**, **170b**, **170c**, and **170d**, sequentially stacked. For example, the first contact layer **170a** may include undoped polycrystalline silicon, the second contact layer **170b** may include doped polycrystalline silicon, the third contact layer **170c** may include a silicide material, and the fourth contact layer **170d** may include a metal. However, depending on example embodiments, the number of layers and type of materials of the contact patterns **170** may vary.

[0055] The semiconductor device **100** may further include upper insulating patterns **175** disposed between the contact patterns **170** (in the X-direction). Each of the upper insulating patterns **175** may extend vertically (in the Z-direction) and may contact the insulating structure **160** and/or the upper capping layer **126**. The upper insulating patterns **175** may spatially separate the contact patterns **170** and electrically insulate the same. For example, the upper insulating patterns **175** and the contact patterns **170** may be alternately arranged in the X-direction.

[0056] The information storage structures **180** may include first electrodes **182** electrically connected to the contact patterns **170**, second electrodes **186** on (overlapping or covering) the first electrodes **182**, and a dielectric layer **184** between the first electrodes **182** and the second electrodes **186**.

[0057] In an example embodiment, the information storage structures **180** may be capacitors storing information in a DRAM. For example, the dielectric layer **184** of the information storage structures **180** may be a capacitor dielectric layer of a DRAM, and the dielectric layer **184** may include a high- κ dielectric material, silicon oxide, silicon nitride, silicon oxynitride, and/or a combination thereof.

[0058] Depending on example embodiments, the information storage structures **180** may be structures storing DRAM and other memory information. For example, the dielectric

layer **184** of the information storage structure **180** may be a capacitor dielectric layer of a ferroelectric memory (Fe-RAM). In this case, the dielectric layer **184** may be a ferroelectric layer that can record data using a polarization state. In example embodiments, the dielectric layer **184** may include a lower dielectric layer including silicon oxide and/or a high- κ dielectric and a ferroelectric layer disposed on the lower dielectric layer.

[0059] FIGS. 4 to 19 are cross-sectional views shown according to a process sequence to illustrate a method of manufacturing a semiconductor device according to an example embodiment.

[0060] Referring to FIG. 4, a mask layer **M** may be formed on the semiconductor substrate **10**. In an example embodiment, the semiconductor substrate **10** may be a silicon on insulator (SOI) substrate. The semiconductor substrate **10** may include a lower semiconductor layer **11**, an insulating layer **12**, and an upper semiconductor layer **13**. For example, the upper and lower semiconductor layers **11** and **13** may include monocrystalline silicon. In some example embodiments, the semiconductor substrate **10** may be a bulk silicon substrate.

[0061] Referring to FIG. 5, back gate trenches **T1** may be formed within the semiconductor substrate **10**. The back gate trenches **T1** may vertically penetrate (extend in the Z-direction) the mask layer **M**, the upper semiconductor layer **13**, and the insulating layer **12**, and expose an upper surface of the lower semiconductor layer **11**. The back gate trenches **T1** may extend in the Y-direction and may be spaced apart from each other in the X-direction.

[0062] A dielectric material layer **122p** may be formed conformally along inner walls of the back gate trenches **T1**. The dielectric material layer **122p** may be on (e.g., cover) the side surfaces of the mask layer **M**, the upper semiconductor layer **13**, and the insulating layer **12**, and may be on (e.g., cover) the upper surface of the lower semiconductor layer **11**.

[0063] Referring to FIG. 6, a back gate electrode **124** and a preliminary capping layer **126p** may be formed on the dielectric material layer **122p**. The back gate electrode **124** may fill lower portions of the back gate trenches **T1**, and the preliminary capping layer **126p** may be disposed on the back gate electrode **124** (may fill upper portions of the back gate trenches **T1**). In an example embodiment, the back gate electrode **124** may include metal nitride, such as TiN, or polycrystalline silicon.

[0064] The preliminary capping layer **126p** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, and/or a combination thereof. In an example embodiment, the preliminary capping layer **126p** may include silicon oxide.

[0065] Referring to FIG. 7, gate trenches **T2** may be formed within the upper semiconductor layer **13**. The gate trenches **T2** may be formed by anisotropically etching the upper semiconductor layer **13** to expose the upper surface of the insulating layer **12**. The gate trenches **T2** may extend in the Y-direction and may be spaced apart from each other in the X-direction. The upper semiconductor layer **13** patterned by the etching process of the gate trenches **T2** may be referred to as a channel layer **140**. In some embodiments, the back gate trenches **T1** and the gate trenches **T2** may be spaced apart from each other in the X-direction. For example, the back gate trenches **T1** and the gate trenches **T2** may be alternately arranged in the X-direction.

[0066] After the etching process of the gate trenches **T2**, the channel layer **140** may be further patterned in the X-direction. As shown in FIG. 1, the patterned channel layers **140** may be spaced apart from each other in the Y-direction along the back gate electrodes **124**.

[0067] Lower insulating patterns **130** may be formed between the channel layers **140**. The lower insulating patterns **130** may be on (may contact) lower regions of the side surfaces of the channel layers **140**. The lower insulating patterns **130** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, and/or a combination thereof. In an example embodiment, the lower insulating patterns **130** may include silicon oxide.

[0068] Referring to FIG. 8, a dielectric material layer **150p** may be formed conformally along inner walls of the gate trenches **T2**. The dielectric material layer **150p** may be on (e.g., cover) the mask layer **M** (e.g., a side surface and an upper surface (an upper end) of the mask layer **M**), the dielectric material layer **122p** (e.g., an upper surface (an upper end) of the dielectric material layer **122p**), the preliminary capping layer **126p** (e.g., an upper surface (an upper end) of the preliminary capping layer **126p**), the lower insulating pattern **130** (e.g., an upper surface of the lower insulating pattern **130**), and the channel layer **140** (e.g., a side surface of the channel layer **140**).

[0069] Referring to FIG. 9, word lines **152**, a sacrificial layer **160p**, and a gate capping layer **162** may be formed. The word lines **152** may be formed on the dielectric material layer **150p** within the gate trenches **T2**. For example, the word line **152** may be on a side surface and an upper surface of the dielectric material layer **150p** exposed by the gate trenches **T2**. The word lines **152** may be formed by depositing a conductive material on the dielectric material layer **150p**, and anisotropically etching the conductive material. As shown in FIG. 1, the word lines **152** may extend in the Y-direction along the back gate electrodes **124**, and may be spaced apart from each other in the X-direction.

[0070] The gate capping layer **162** may be formed on the word lines **152**. In an example embodiment, the gate capping layer **162** may be formed by forming an insulating material on the conductive material, and anisotropically etching the insulating material together with the conductive material. For example, the gate capping layer **162** may be on an upper surface of the word line **152** and a side surface of the dielectric material layer **150p**.

[0071] The sacrificial layer **160p** may be formed between the word lines **152** and the gate capping layers **162** (in the X-direction). The sacrificial layer **160p** may include a material having etch selectivity with that of the gate capping layer **162**. For example, the gate capping layer **162** may include silicon nitride, and the sacrificial layer **160p** may include silicon oxide. In some example embodiments, the sacrificial layer **160p** may include the same material as the gate capping layer **162** and may be formed integrally with the gate capping layer **162**.

[0072] A planarization process may be performed to remove the mask layer **M** and expose upper surfaces of the channel layers **140**. Upper portions of the dielectric material layer **122p** and the dielectric material layer **150p** may be removed to form a back gate dielectric layer **122** and a gate dielectric layer **150**, respectively.

[0073] Referring to FIG. 10, the sacrificial layer **160p** may be removed and the word lines **152** may be exposed. For example, the sacrificial layer **160p** between the gate capping

layers **162** may be selectively removed by a wet etching process. In an example embodiment, the preliminary capping layer **126p** may also be removed through the etching process, and the back gate electrodes **124** may be exposed. In some example embodiments, the preliminary capping layer **126p** may include a material having etch selectivity with the sacrificial layer **160p**, and the material may not be removed. In an example embodiment, upper portions of the back gate dielectric layer **122** and the gate dielectric layer **150** may be (partially) etched.

[0074] Referring to FIG. 11, a liner material layer **164p** may be formed. The liner material layer **164p** may be on (e.g., cover) a side surface (and/or an upper surface) of the back gate dielectric layer **122**, and an upper surface of the back gate electrode **124**. The liner material layer **164p** may also be on (e.g., cover) a side surface of the word line **152**, a side surface (and/or an upper surface) of the gate capping layer **162**, and an upper surface of the gate dielectric layer **150**. The liner material layer **164p** may include, for example, silicon nitride and may cover the word lines **152** to prevent the word lines **152** from being oxidized. An upper surface of an element may include an uppermost surface of the element and an upper surface of the element lower than the uppermost surface of the element. For example, the liner material layer **164p** may be on the uppermost surface of the gate dielectric layer **150** and on an upper surface of the gate dielectric layer **150** lower than the uppermost surface of the gate dielectric layer **150**.

[0075] Referring to FIG. 12, a capping material layer **166p** may be formed. The capping material layer **166p** may be on (e.g., cover) upper surfaces of the channel layers **140**, and (at least partially) fill a space between the back gate dielectric layers **122**. A portion of the capping material layer **166p** may be disposed between the gate capping layers **162**, and may define an air gap AG. The capping material layer **166p** may partially fill a space between the adjacent gate capping layers **162** in the X-direction. For example, the air gap AG may be disposed between adjacent word lines **152** (in the X-direction), and between the capping material layer **166p** (which will become the capping pattern **166** in later processes) and the liner material layer **164p** (which will become the insulating liner **164**) (in the Z-direction).

[0076] Referring to FIG. 13, the liner material layer **164p** and the capping material layer **166p** may be etchbacked to form an insulating liner **164** and a capping pattern **166**. The capping pattern **166** may define an upper limit of the air gap AG, and the insulating liner **164** may define a lateral limit (in the X-direction) and a lower limit of the air gap AG. A portion of the capping material layer **166p** (at least partially) filling the space between the back gate dielectric layers **122** (e.g., between the back gate liner **127**) may be referred to as an upper capping layer **126**. The upper surfaces of the channel layers **140** may be exposed through the etch-back process.

[0077] Referring to FIG. 14, a contact material layer **170'** may be formed on the channel layers **140**. The contact material layer **170'** may include a first contact material layer **170a'**, a second contact material layer **170b'**, a third contact material layer **170c'**, and a fourth contact material layer **170d'**, sequentially stacked.

[0078] Referring to FIG. 15, the contact material layer **170** may be patterned to form a contact pattern **170**. The contact pattern **170** may include a first contact pattern **170a**, a second contact pattern **170b**, a third contact pattern **170c**,

and a fourth contact pattern **170d**, sequentially stacked. The contact pattern **170** may be electrically connected to the channel layer **140**.

[0079] Upper insulating patterns **175** may be formed between the contact patterns **170**. The upper insulating patterns **175** may be formed by patterning the contact material layer **170'** and filling the same with an insulating material. The upper insulating patterns **175** may electrically separate the contact patterns **170** from each other. For example, the upper insulating patterns **175** and the contact patterns **170** may be alternately arranged in the X-direction.

[0080] An information storage structure **180** including first electrodes **182**, a dielectric layer **184**, and a second electrode **186** may be formed on the contact patterns **170**. The first electrodes **182** may contact the fourth contact patterns **170d** of the contact patterns **170**.

[0081] Referring to FIG. 16, the resulting structure of FIG. 15 may be turned over so that the information storage structure **180** faces downwardly of the lower semiconductor layer **11** and a grinding process may be performed. Through the grinding process, the lower semiconductor layer **11** and the insulating layer **12** may be removed, and the lower insulating pattern **130** and the channel layer **140** may be exposed. The back gate dielectric layer **122** and the back gate electrode **124** may also be partially etched to be coplanar with the channel layer **140**.

[0082] Referring to FIG. 17, a portion of the back gate electrode **124** may be removed, and a lower capping layer **128** may be formed in a space in which the portion of the back gate electrode **124** has been removed. The back gate electrode **124** may be disposed between the upper capping layer **126** and the lower capping layer **128** and may not be exposed.

[0083] Referring to FIG. 18, a cleaning process may be performed. Through the cleaning process, an oxide film formed on the channel layers **140** may be removed. Through the cleaning process, the back gate dielectric layer **122** and the lower insulating pattern **130** may be partially etched and side surfaces (of upper regions) of the channel layers **140** may be exposed.

[0084] Referring to FIGS. 2 and 19, the bit line structure **110** and the lower insulating layer **101** may be formed on the channel layers **140** so that the semiconductor device **100** may be manufactured. The bit line structure **110** may include a third conductive pattern **110c**, a second conductive pattern **110b**, and a first conductive pattern **110a**, sequentially stacked on the channel layers **140**.

[0085] In an example embodiment, a peripheral circuit structure including peripheral circuit elements electrically connected to at least one of the bit line structures **110** may be disposed on the lower insulating layer **101**.

[0086] FIGS. 20 to 25 are cross-sectional views of semiconductor devices according to example embodiments.

[0087] Referring to FIG. 20, an insulating structure **160** of a semiconductor device **100a** may include a gate capping layer **162**, a capping pattern **166a**, an air gap AG, and an insulating liner **164**. The capping pattern **166a** and the insulating liner **164** may define the air gap AG. In an example embodiment, a lower surface of the capping pattern **166a** defining an upper limit of the air gap AG may be concave. The capping pattern **166a** may be concave toward the upper insulating patterns **175** in the Z-direction.

[0088] Referring to FIG. 21, an insulating structure **260** of a semiconductor device **100b** may include a gate capping

layer 162, an insulating liner 164, a capping pattern 166, and a low- κ dielectric material 266 disposed on the insulating liner 164. For example, the low- κ dielectric material 266 may be disposed on an inner surface 164a (and/or an outer surface 164b) of the insulating liner 164. In an example embodiment, the low- κ dielectric material 266 may be disposed in the core region, and completely fill a space between the insulating liners 164 in the X-direction. The low- κ dielectric material 266 may be formed between the word lines 152 during the process of forming the capping material layer 166p described with reference to FIG. 12. An upper surface of the low- κ dielectric material 266 may be coplanar with an upper surface of the insulating liner 164 and an upper surface of the gate capping layer 162.

[0089] Referring to FIG. 22, a back gate structure 120 of a semiconductor device 100c may include a back gate dielectric layer, a back gate electrode 124, an upper capping layer 126c on the back gate electrode 124, and the lower capping layer 128. In an example embodiment, unlike the embodiment shown in FIG. 3, a back gate liner 127 may be omitted. For example, the upper capping layer 126c may be on (may contact) an upper surface of the back gate electrode 124 and a side surface of the back gate dielectric layer 122 (without the back gate liner 127 therebetween).

[0090] In the wet etching process described with reference to FIG. 10, a preliminary capping layer 126p may include a material having etch selectivity with the sacrificial layer 160p and may not be removed. For example, the preliminary capping layer 126p may include silicon nitride. In the process of forming a liner material layer 164p described with reference to FIG. 11, the liner material layer 164p may be formed on the preliminary capping layer 126p and may not contact the back gate electrode 124. The preliminary capping layer 126p after the etch-back process described with reference to FIG. 13 may be referred to as an upper capping layer 126c.

[0091] Referring to FIG. 23, a back gate structure 120 of a semiconductor device 100d may include a back gate dielectric layer 122, a back gate electrode 124, an upper capping layer 126, a back gate liner 127, and a lower capping layer 128. In the wet etching process described with reference to FIG. 10, at least one of an upper portion of the back gate dielectric layer 122 and an upper portion of the gate dielectric layer 150 may be etched.

[0092] In an example embodiment, a horizontal width (in the X-direction) of the back gate dielectric layer 122 may decrease upwardly. An upper end of the back gate dielectric layer 122 may be located on a level lower than an upper surface of the channel layer 140. A horizontal width (in the X-direction) of the upper capping layer 126 may decrease downwardly. A portion of the upper capping layer 126 may overlap the back gate dielectric layer 122 in a vertical direction (in the Z-direction). A portion of the back gate liner 127 may be in contact with the channel layer 140.

[0093] A semiconductor device 100d may further include an insulating layer 163d disposed on (e.g., in contact with) a side surface of the insulating structure 160 and an upper surface (e.g., the uppermost surface) of the gate dielectric layer 150. The insulating layer 163d may contact a side surface of the channel layer 140 and a side surface of the gate capping layer 162. The insulating layer 163d may also contact an upper surface of the gate dielectric layer 150. The insulating layer 163d may include, for example, silicon nitride.

[0094] Referring to FIG. 24, a back gate structure 120 of a semiconductor device 100e may include a back gate dielectric layer 122, a back gate electrode 124, a first upper capping layer 126e1 on the back gate electrode 124, second upper capping layers 126e2 disposed both side surfaces (opposite side surfaces in the X-direction) of the first upper capping layer 126e1, back gate liners 127e on (e.g., covering) side surfaces and lower surfaces of the second upper capping layers 126e2, and a lower capping layer 128. The back gate liners 127e may contact a side surface of the first upper capping layer 126e1. The first upper capping layer 126e1, the second upper capping layers 126e2, and the back gate liners 127e may include, for example, silicon nitride. The semiconductor device 100e may include the insulating layer 163d that is included in the semiconductor device 100d in FIG. 23.

[0095] Referring to FIG. 25, an insulating structure 160 of a semiconductor device 100f may include a capping pattern 366, an insulating liner 364, and an air gap AG. In an example embodiment, unlike the embodiment shown in FIG. 3, a gate capping layer 162 may be omitted. For example, the gate capping layer 162 may include the same material as the sacrificial layer 160p, and in the wet etching process described with reference to FIG. 10, the gate capping layer 162 can be removed together with the sacrificial layer 160p.

[0096] The insulating liner 364 may be in contact with upper and side surfaces of the word lines 152, and a side surface of the gate dielectric layer 150. A portion of the capping pattern 366 may overlap the word lines 152 in a vertical direction (in the Z-direction). A lower end of the capping pattern 366 may be located on a level lower than the upper surfaces of the word lines 152. The capping pattern 366 may have a horizontal width (in the X-direction) greater than the horizontal width (in the X-direction) of the air gap AG.

[0097] As set forth above, according to example embodiments of the technical idea of the present inventive concept, since an air gap or a low- κ dielectric material is disposed between word lines, coupling noise between adjacent word lines may be reduced or prevented. Accordingly, it is possible to prevent leakage current from occurring in an unselected transistor when a semiconductor device is operating.

[0098] The various and advantageous advantages and effects of the present inventive concept are not limited to the above description, and may be more easily understood in the course of describing the specific embodiments of the present inventive concept.

[0099] While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concept as defined by the appended claims.

What is claimed is:

1. A semiconductor device, comprising:
 - a bit line structure;
 - back gate structures on the bit line structure;
 - a first channel layer and a second channel layer between the back gate structures in a first direction, wherein the first channel layer and the second channel layer extend in a second direction;
 - a first word line and a second word line between the first channel layer and the second channel layer in the first direction, wherein the first word line is adjacent to the

first channel layer, and the second word line is adjacent to the second channel layer; and
 an insulating structure on a side surface of the first word line, a side surface of the second word line, an upper surface of the first word line, and an upper surface of the second word line,

wherein the insulating structure includes

a core region between the first word line and the second word line in the first direction; and

an insulating liner between the core region and the first word line in the first direction and between the core region and the second word line in the first direction, wherein the side surface of the first word line and the side surface of the second word line face each other in the first direction,

wherein the insulating liner extends to a level higher than those of the first word line and the second word line, wherein the first direction is parallel with a lower surface of the bit line structure, and

wherein the second direction is perpendicular to the lower surface of the bit line structure.

2. The semiconductor device of claim 1, wherein the insulating structure further comprises a first gate capping layer on the upper surface of the first word line and a second gate capping layer on the upper surface of the second word line, and

wherein the insulating liner is between the core region and the first gate capping layer in the first direction and between the core region and the second gate capping layer in the first direction.

3. The semiconductor device of claim 2, wherein the insulating structure further comprises a capping pattern on the core region, and

wherein the insulating liner is between the capping pattern and the first gate capping layer in the first direction and is between the capping pattern and the second gate capping layer in the first direction.

4. The semiconductor device of claim 3, wherein a lower surface of the capping pattern is convex toward the core region.

5. The semiconductor device of claim 1, wherein the core region includes an air gap.

6. The semiconductor device of claim 1, further comprising:

a gate dielectric layer below a lower surface of the insulating structure,

wherein the gate dielectric layer is between the first word line and the first channel layer in the first direction and between the second word line and the second channel layer in the first direction, and

wherein the insulating liner is between the core region and the gate dielectric layer in the second direction.

7. The semiconductor device of claim 1, wherein each of the back gate structures comprises a back gate electrode, an upper capping layer on the back gate electrode, and a back gate liner on a lower surface and a side surface of the upper capping layer.

8. The semiconductor device of claim 7, wherein the back gate liner comprises a same material as that of the insulating liner.

9. The semiconductor device of claim 7, wherein a horizontal width of the upper capping layer in the first direction decreases as the upper capping layer extends toward the back gate electrode in the second direction.

10. The semiconductor device of claim 9, wherein each of the back gate structures further comprises a back gate dielectric layer on a side surface of the back gate electrode and the side surface of the upper capping layer, and

wherein a horizontal width of the back gate dielectric layer in the first direction decreases as the back gate dielectric layer extends farther from the lower surface of the bit line structure.

11. The semiconductor device of claim 1, wherein the insulating structure further comprises a capping pattern on the core region, and

wherein a lower surface of the capping pattern is concave in the second direction.

12. The semiconductor device of claim 1, wherein the core region comprises a low-k dielectric material having a dielectric constant lower than that of silicon oxide.

13. The semiconductor device of claim 1, wherein each of the back gate structures comprises a back gate electrode, a first upper capping layer on the back gate electrode, a back gate dielectric layer on a side surface of the back gate electrode, and second upper capping layers on opposite sides of the first upper capping layer in the first direction.

14. The semiconductor device of claim 1, wherein the insulating structure further comprises a capping pattern on the core region,

wherein a portion of the capping pattern overlaps the first word line and the second word line in the second direction,

wherein a lower end of the capping pattern is closer than the upper surface of the first word line and the upper surface of the second word line to the lower surface of the bit line structure in the second direction, and

wherein the insulating structure is in contact with the side surface of the first word line, the side surface of the second word line, the upper surface of the first word line, and the upper surface of the second word line.

15. The semiconductor device of claim 14, wherein the insulating liner is on the upper surface of the first word line and the upper surface of the second word line.

16. A semiconductor device, comprising:

a bit line structure;

a channel layer on the bit line structure, wherein the channel layer extends in a first direction;

a back gate structure facing a first side surface of the channel layer in a second direction;

a word line facing a second side surface of the channel layer that is opposite to the first side surface of the channel layer in the second direction;

a gate dielectric layer between the channel layer and the word line in the second direction; and

an insulating structure on the word line,

wherein the gate dielectric layer includes a vertical portion in contact with a first side surface of the word line and a side surface of the insulating structure, and a horizontal portion in contact with a lower surface of the word line and a lower surface of the insulating structure,

wherein the insulating structure includes

a core region spaced apart from the word line; and

an insulating liner between the core region and a second side surface of the word line in the second direction, and extending between the horizontal portion and the core region in the first direction,

wherein the first side surface of the word line is opposite to the second side surface of the word line in the second direction,

wherein the first direction is perpendicular to a lower surface of the bit line structure, and

wherein the second direction is parallel with the lower surface of the bit line structure.

17. The semiconductor device of claim **16**, wherein the insulating structure further comprises a capping pattern in contact with an upper region of an inner surface of the insulating liner, and

wherein the core region includes an air gap.

18. The semiconductor device of claim **16**, wherein the back gate structure further comprises a back gate electrode and a back gate dielectric layer between the back gate electrode and the channel layer in the second direction, and

wherein a lower surface of the back gate dielectric layer is farther than a lower surface of the channel layer from the lower surface of the bit line structure.

19. The semiconductor device of claim **16**, further comprising:

an insulating pattern between the bit line structure and the gate dielectric layer in the first direction.

20. A semiconductor device, comprising:

a bit line structure;

back gate structures on the bit line structure;

a first channel layer and a second channel layer between the back gate structures in a first direction, wherein the first channel layer and the second channel layer extend in a second direction;

a first word line and a second word line between the first channel layer and the second channel layer in the first direction, wherein the first word line is adjacent to the first channel layer, and the second word line is adjacent to the second channel layer;

an insulating structure in contact with a side surface of the first word line, a side surface of the second word line,

an upper surface of the first word line, and an upper surface of the second word line;

a gate dielectric layer between the first channel layer and the first word line in the first direction, between the first channel layer and a first upper region of the insulating structure in the first direction, between the second channel layer and the second word line in the first direction, and between the second channel layer and a second upper region of the insulating structure in the first direction;

contact patterns electrically connected to the first channel layer and the second channel layer; and

an information storage structure on the contact patterns, wherein the insulating structure includes a core region spaced apart from the first word line and second word line; and

an insulating liner between the side surface of the first word line and the core region and between the side surface of the second word line and the core region, and extending to a level higher than those of the first word line and the second word line,

wherein the core region has a dielectric constant lower than that of silicon oxide,

wherein the first upper region of the insulating structure is on the first word line,

wherein the second upper region of the insulating structure is on the second word line,

wherein the side surface of the first word line and the side surface of the second word line face each other in the first direction,

wherein the first direction is parallel with a lower surface of the bit line structure, and

wherein the second direction is perpendicular to the lower surface of the bit line structure.

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